

Factor models for Chinese A-shares^{*}

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Abstract:

We compare the performance of commonly employed asset pricing models on a large, liquid, but mostly segmented Chinese A-shares equity market. When restricting ourselves to factor models developed for the U.S. equity market, the q-factor model performs well. However, it is outperformed by a modified Fama-French six-factor model and by a four-factor asset pricing model tailored to the Chinese A-shares market. A data-driven method results in a seven-factor model, however the ranking of asset pricing models changes when we incorporate transaction costs. Both direct and data-driven model comparison methods now lead to a three-factor model comprising a market, size, and earnings-based value factor.

Keywords: Anomalies, Asset pricing, China, Equity markets, Emerging markets, Factor models, Investing

JEL code: C58, D53, G11, G12, G15

1. Introduction

China is the second-largest economy in the world. In 2000, its gross domestic product was already 36% of that of the U.S. and steadily increased to 116% in 2020.¹ Furthermore, the Chinese equity market has evolved into the second-largest (after the U.S.) in terms of market capitalization and trading volume.² Despite its economic importance, Chinese A-shares have been mostly segmented from other equity markets in the world (see Hu, Pan, and Wang, 2021), and firms tend to have different corporate governance features, leading to different agency problems (see Jiang and Kim, 2020).³ This makes the A-share market an excellent out-of-sample environment to test which of the factors drive the cross-section of expected returns; see, e.g., Hsu, Viswanathan, Wang, and Wool (2018), Gu, Jiang, and Xu (2019), Blitz, Hanauer, and van Vliet (2021), Jansen, Swinkels, and Zhou (2021), Yin and Yang (2022), and Hou, Qiao, and Zhang (2023). A study focusing on the China A-share market adds to studies that investigate a large sample of developed or emerging countries, because it has been largely segmented from those other markets. For markets that are largely financially integrated, it is much more likely to find the same patterns in stock returns than in markets with a completely different set of investors.⁴

Cochrane (2011), Harvey, Liu, and Zhu (2016), and Hou, Xue, and Zhang (2020) raise concerns about the burgeoning 'zoo of factors' that has emerged in the asset pricing literature over the past decades. To determine whether factors are real or spurious, one can examine their existence in international markets that are imperfectly integrated with U.S. equity markets. While international research is relatively scarce (cf., Karolyi, 2016), recent studies that have examined the existence of factors in other developed, emerging, and frontier markets; see, e.g., Jacobs and Müller (2020), Hanauer and Lauterbach (2019), and De Groot, Pang, and Swinkels (2012), respectively. In addition, the apparent existence of a large number of anomalous return patterns has spurred research into new asset pricing models that better account for these anomalies relative to the Capital Asset Pricing Model or the popular Fama and French (1993) three-factor model.

¹ Source: IMF World Economic Outlook (October 2020). Numbers in the main text are purchasing power parity adjusted (i.e. in 'international dollars'). Unadjusted numbers (i.e. in current dollars) would be 12% and 71%, respectively.

² Source: World Bank Development Indicators CM.MKT.LCAP.CD and CM.MKT.TRAD.CD for 2019, respectively.

³ The A-share market contains the shares of companies from China traded in RMB on the stock exchanges of Shanghai and Shenzhen. The B-share market is the group of stocks listed on the same exchanges, but trade in USD or HKD instead. The H-share market is the group of Chinese stocks that are listed on the Hong Kong Stock Exchange and therefore trade in HKD. The A-share market is by far the largest measured by market value. For more details on the development of these markets, see Weber and Zhang (2012).

⁴ See also Lucey et al. (2018) for an overview and future directions in international financial integration research.

Competing factor models are, among others, the five- and six-factor model by Fama and French (2015, 2018), the q-factor model by Hou, Xue, and Zhang (2015), and the mispricing model by Stambaugh and Yuan (2017). Several recent papers compare the performance of asset pricing models for the cross-section of U.S. stock returns; see, e.g., Barillas and Shanken (2018), Fama and French (2018), Barillas, Kan, Robotti, and Shanken (2020), and Hou, Mo, Xue, and Zhang (2019). International asset pricing model comparisons are conducted by Fletcher (2018, 2019), Hanauer (2020), and Huber, Jacobs, Müller, and Preissler (2023).

Several studies examine whether the Fama and French (1993) three-factor model is able to price the cross-section of stock returns of Chinese A-shares; see, e.g., Wang and Xu (2004), Eun and Huang (2007), Wang and Di Iorio (2007), Huang, Yang, and Zhang (2013), Xu and Zhang (2014), Xie and Qu (2016), Hu, Chen, Shao, and Wang (2019), and Liu, Stambaugh, and Yuan (2019). A few studies investigate the performance of the profitability and investment factors of the five-factor model by Fama and French (2015). For example, Guo, Zhang, Zhang, and Zhang (2017), Lin (2017), and Huang (2019) find that the five-factor model is better able to price the cross-section of Chinese stock returns than the three-factor model. This improvement is mainly due to the addition of a profitability factor, while the investment factor is relatively weak. A comprehensive comparison between competing asset pricing models has not yet been conducted for the Chinese A-shares market and this paper aims to fill this gap.

Our main findings are summarized as follows. First, we find that the conventional three- and five-factor models from Fama and French (1993, 2015) poorly price the cross-section of Chinese stock returns. Adding a momentum factor and replacing the original book-to-market and operating profitability factors with monthly updated book-to-market and return-on-equity factors improves their performance significantly. Second, a parsimonious four-factor model tailored to the China A-share market outperforms the existing models developed to price the cross-section of U.S. equity returns. Using a data-driven Bayesian asset pricing framework (Barillas and Shanken, 2018; Chib, Zeng, and Zhao, 2020), we find that a seven-factor model comprising the same four factors together with three additional ones achieves the highest maximum (squared) Sharpe ratio. Third, we account for transaction costs to examine which model spans the achievable frontier, which represents the investment opportunity set attainable for investors. This analysis builds upon Detzel, Novy-Marx, and Velikov (2023) for the U.S. stock market, who argue that arbitrage can only be

expected to eliminate trading opportunities that can be profitably exploited in practice. Failing to account for transaction costs favors factors with large gross alphas and high turnover that might not be profitably traded. After adjusting for transaction costs, the preferred model is a three-factor model comprising a market, size, and monthly updated earnings-based value factor. Lastly, we find that value factors are part of the data-driven asset pricing model, regardless of whether we include or exclude transaction costs. Our empirical evidence thus indicates that value is not a redundant asset pricing factor for China A-shares and that its dismissal by some based on its recent poor performance in the U.S. equity market may be premature.

The contribution of this paper is threefold. First, we compare the pricing ability of various asset pricing models for a large, liquid, but mostly segmented equity Chinese A-share market. This analysis sheds light on which factors are more likely to be universal drivers of risk and return and which factors are more local or transient in nature. This is important, as the recent literature found that some reported stylized facts in equity markets turn out to be US-specific and are not present in international markets. For example, Cakici and Zaremba (2022) do not find evidence of the salience theory in international stock markets that was documented by Cosemans and Frehen (2021) for the US stock market, and Cakici, Fieberg, Metko, and Zaremba (2023) report that anomalies in international markets do not predict market returns, contrary to the US evidence provided by Dong, Li, Rapach, and Zhou (2022). Our work complements this recent strand of literature on the possible generalization of stylized factors across financial markets. Second, we are the first to employ a data-driven method to determine the preferred asset pricing model for the Chinese A-share market and compare these with the pricing ability of models developed and tested using U.S. stock markets. Third, we evaluate which factor model most closely spans the achievable efficient frontier when accounting for transaction costs. Detzel, Novy-Marx, and Velikov (2023) show that accounting for transaction costs materially changes the ranking of competing asset pricing models. Model selection studies that ignore transaction costs favor models that include more frequently updated factors with less persistent characteristics. However, Novy-Marx and Velikov (2016) show that such high turnover factors often do not generate significant net returns despite high gross returns and thus do not yield abnormal returns in real trading. Our paper adds to this literature by studying the implications of transaction costs on the performance of asset pricing models in the A-share market.

Each of these three contributions also adds to the discussion on the existence of the value premium and whether asset pricing models should contain a value factor to price the cross-section of stock returns. Several recent studies, including Fama and French (2015), have suggested that the value premium may be redundant in asset pricing models for the U.S. equity market. The discussion of whether there ever really was a value premium or whether the value factors as currently defined will persist is hotly debated at the moment.⁵

The results presented in this study have important implications for defining a powerful asset pricing model for Chinese A-shares. The seven-factor data-driven asset pricing model is the best choice if the goal is to find the factor model with superior standalone performance over the sample period. However, we cannot rule out that such a data-driven asset pricing model is a result of overfitting. Suppose one looks for an asset pricing model that performs consistently well across international markets. In that case, a modified Fama-French six-factor model with a cash-based operating profitability and a monthly updated HML factor is most suitable. Finally, the preferred asset pricing model from an investor's perspective is the Liu, Stambaugh, and Yuan (2019) three-factor model since it performs best after accounting for transaction costs.

The remainder of this paper is organized as follows. Section 2 contains a description of the data and the factor construction methodology. In Section 3, we describe the methodology to compare factor models and report the empirical results for existing factor models. Section 4 contains the results of the Bayesian data-driven asset pricing framework, while Section 5 deals with the role of transaction costs in comparing asset pricing models. Section 6 concludes.

2. Data and factor construction

2.1. Data

The data used in this paper is obtained from the China Stock Market and Accounting Research (CSMAR) database. From this database, we retrieve stock price data and quarterly financial statement data for all RMB-denominated stocks listed on the Shanghai Stock Exchange and Shenzhen Stock Exchange. This includes Main Board stocks, which are mature stocks similar to

⁵ The recent value debate includes, among others, Arnott, Harvey, Kalesnik, and Linnainmaa (2021), Blitz and Hanauer (2021), Fama and French (2021), Israel, Laursen, and Richardson (2021), and Lev and Srivastava (2022).

those on the New York Stock Exchange, as well as SSE Star Market and SZSE ChiNext stocks, which resonate more closely with Nasdaq stocks. Our sample period is from July 2000 to December 2019. The primary reason for this start date is to ensure a sufficiently broad cross-section of stocks. Liu, Stambaugh, and Yuan (2019) start their analyses in 2000 for the same reason. Table A1 in Appendix A shows that there were less than 1,000 listed firms before 2000. The period up until 1996 is also characterized by extreme returns and volatility, with annual market returns ranging between -30 and 150 percent and market volatility peaking at 134 percent in 1994. The 19.5-year sample period is relatively short, even though the original three-factor model by Fama and French (1993) also used only 29 years of return data. The sample length of our study is roughly two-thirds of theirs.⁶ During our 19.5-year sample period, there were some important regulatory changes, such as the Split-Share Structure Reform. Jansen, Swinkels, and Zhou (2021) show that the performance of most characteristic-sorted portfolios is similar before and after this regulatory change. Carpenter, Lu, and Whitelaw (2021) find that the price informativeness in Chinese equity markets is on par with that in the US after the Split-Share Structure Reform took effect.

Due to evolving accounting standards in China, quarterly financial statements are only available from 2002 onward. Prior to this, we use semi-annual reports.⁷ Similar to Liu, Stambaugh, and Yuan (2019), we make use of statement release dates to match accounting data extracted from these reports with returns data. In case no release data is available, we discard all interim reports but retain the annual reports.⁸ For the latter, we assume a statement release date of 1 May because all listed firms are required to disclose their annual report before the end of April. This way, we ensure that no forward-looking information is used in portfolio construction.

⁶ Ledoit, Wolf, and Zhou (2019) develop an efficient sorting procedure that combines information from simple sorting on firm characteristics and the covariance matrix. Jansen, Swinkels, and Zhou (2023) apply this efficient sorting technique on the period after the Split-Share Structure Reform and find that t-statistics are on average double, leading to improved test power for factor returns over an even shorter sample period.

⁷ It is necessary to adjust the magnitude of the flow variables. In the immediate period following 2002, which is the period when statements become quarterly, the period over which the flow variables are calculated differs between firms. For instance, the latest-available accounting statement from one firm in 2002 may be its previous-year annual report, while another firm may have already released a quarterly statement. We address this by multiplying all semi-annual flow data by two and quarterly flow data by four.

⁸ All interim income statement and cash flow statement data, i.e., all flow variables, from CSMAR correspond to cumulative year-to-date values. For example, the quarter three revenue in a particular year corresponds to the sum of the revenue over the first three quarters of that year. This induces a seasonality in the accounting data and invalidates cross-sectional comparison. More specifically, using this data will result in cross-sectionally comparing values calculated over different period lengths with each other. In the most extreme case, one would compare the fourth quarter revenue of a firm, which is the annual revenue, with the first quarter revenue of another firm. To correct for this, we take the first-difference of any subsequent values within an accounting year. This correction ensures that the period over which flow variables are calculated matches exactly with the period of the financial statement. The non-availability of quarterly reports in the first years of our sample makes it challenging to include the Daniel et al. (2020) behavioral asset pricing model for our comparison, as we would lose another two years of our already relatively short sample period.

We apply a series of filters to exclude stocks with unreliable data or abnormal behavior from our portfolios. Following Liu, Stambaugh, and Yuan (2019), we require stocks to have been (i) listed for at least six months, (ii) suspended for less than five trading days in the past month, and (iii) suspended for less than 120 trading days in the past year.⁹ The latter two restrictions are particularly important as they reduce the impact of frequent and/or prolonged trading suspensions on our findings. Trading suspensions occurred regularly over our sample period.¹⁰ Jansen, Swinkels, and Zhou (2021) show that the number of suspensions was negligible prior to 2006 and after 2019. However, more than 20% of listed stocks were suspended from trading in 2007 and 2016, and the share of suspended stocks was between 5% and 10% for the period in between. Tian (2019) documents that failing to properly account for trading suspensions leads to markedly different excess factor returns.

Following Hsu et al. (2018) and Lin (2019), we exclude stocks with a 'special treatment' or 'particular transfer' status. These stocks are subject to trading restrictions and additional controls, such as a maximum daily price change of five percent. Lastly, due to the constrained and costly initial public offering (IPO) process in China, Liu, Stambaugh, and Yuan (2019) show that small stocks have a substantial shell value as potential reverse merger candidates. They show that the market value and return variation of the smallest 30% of stocks is largely driven by their shell component rather than fundamentals. To avoid shell contamination, we follow their recommendation and eliminate microcaps from our sample, the bottom 30% of stocks ranked by market value. By excluding these smallest stocks from our sample, we also address the critique by Hou, Xue, and Zhang (2020) that many apparent anomalies documented in the U.S. disappear once thinly traded microcap stocks are excluded from the sample. Table A2 in Appendix A shows that our results are consistent with the existing literature in the sense that they become somewhat stronger when microcaps are included.

⁹ Strictly speaking Liu, Stambaugh, and Yuan (2019) eliminate stocks with less than 120 trading records in the past year and 15 trading records in the past month. Our filter is equal to theirs when assuming 20 trading days in a month and 240 days in a year.

¹⁰ For example, China Vanke Co. was suspended for 6 months and plunged with the maximum daily amount of 10% on the day it resumed trading: <https://www.bloomberg.com/news/articles/2016-07-04/china-vanke-s-a-shares-plunge-10-after-six-month-suspension>. Another example is Qinghai Salt Lake Co., who reemerged after a trading suspension of 15 months, after which its shares quadrupled in value: <https://www.yicaiglobal.com/news/qinghai-salt-lake-quadruples-as-potash-giant-resumes-trading-after-15-month-suspension>

2.2. Factor construction

Our set of potential asset pricing factors for the Chinese A-share market is motivated by recent advances in the asset pricing literature for the U.S. and other developed equity markets. In this section, we describe the construction and definition of each of these factors. The factor series used in this paper will be made publicly available to facilitate the replication or extension of our research. The description of the associated factor models follows in Section 3.1.

As considerable differences in design choices exist in constructing asset pricing factors (see, e.g., Soebhag, Van Vliet, and Verwijmeren, 2023), we apply a common set of construction rules for all non-market factors to improve comparability. As in Fama and French (1993), the factors are constructed using independent 2×3 sorts on size and each respective factor characteristic. At the end of each month, stocks are split into two groups, small and big, based on the median market capitalization of all stocks in our investable universe, i.e., after applying our data filters.¹¹ Independently, for all factor characteristics but size, stocks are also split into three groups, low, middle, and high, using the 30th and 70th percentiles as breakpoints. The stocks in the long portfolios are associated with high expected returns, while the short portfolio stocks are associated with low expected returns. For each factor, we take the intersection of these sorts to form monthly market capitalization-weighted portfolio returns. Except for market and size, the factors are calculated as the average return of the difference between the small and big long portfolios and the small and big short portfolios. The size factor is constructed to be orthogonal to all other factors by calculating the average return of the difference between all 33 [11 factors $\times$ (1 high, 1 middle, and 1 low)] small stock portfolios and all 33 big stock portfolios. This procedure follows Fama and French (2015), who also compute a multidimensional size factor when extending their three-factor model to a five-factor model. Furthermore, this procedure allows us to use the same size factor in all models. Portfolio returns are calculated monthly, but rebalancing frequency differs depending on the factor. Some portfolios are updated at the end of each month, while others are updated annually at the end of June.¹²

¹¹ The size split uses breakpoints using all stocks rather than only stocks from the Main Board. The latter is common in studies on the U.S. stock market, where NYSE size breakpoints are used, and Nasdaq stocks are assigned to size portfolios later. For China A-shares we verified that there is virtually no difference between the two in Table A2 in Appendix A. Therefore, we opt for the most straightforward method. This is consistent with the findings of Xu and Zhang (2014).

¹² For annually rebalanced factors, the size breakpoints are also updated annually. For monthly rebalanced factors, the size allocation is also updated monthly.

We provide brief definitions of the 13 factors below. Exact variable definitions and construction are detailed in Appendix B.

- *RMRF*: The market factor is the monthly rebalanced value-weighted portfolio of all investable (after applying our filters) A-shares in CSMAR minus the one-year RMB lump-sum deposit rate, which is updated monthly. This risk-free rate is commonly used as a proxy for the Chinese risk-free rate in this line of literature; see, e.g., Hsu et al. (2018) and Liu, Stambaugh, and Yuan (2019).
- *SMB*: Size is measured by market capitalization. The small and big portfolios are the result of double sorting on each of the non-market factors below. Note that the 30% smallest stocks are already filtered out and are not part of our small size portfolio, consistent with Liu, Stambaugh, and Yuan (2019).
- *HML*: Value is measured by the ratio of the firm's book value of equity to its market capitalization. It is updated annually in June. This is the 'standard' value factor introduced by Fama and French (1993).
- *HML_m*: Same as HML but updated monthly using the latest month-end total market capitalization and the previous year-end book equity; see Asness and Frazzini (2013), Barillas and Shanken (2018), and Barillas et al. (2020).
- *HML_{EP_m}*: Value is measured by the ratio of the firm's earnings to its market capitalization. It is updated monthly using the latest month-end total market capitalization and the most recently reported quarterly earnings. This earnings-based value factor is referred to as Value-Minus-Growth (VMG) in Liu, Stambaugh, and Yuan (2019).
- *RMW_{OP}*: Profitability measured by operating profits after interest to book equity. It is updated annually in June, as in Fama and French (2015).
- *RMW_{CbOP}*: Profitability measured by cash-based operating profits to total assets. It is updated annually in June, as in Ball, Gerakos, Linnainmaa, and Nikolaev (2016).
- *RMW_{ROE_m}*: Profitability measured by return-on-equity, as quarterly earnings divided by book equity. It is updated monthly using the most recent values, as in Hou, Xue, and Zhang (2015).
- *PERF_m*: Performance is a composite factor comprising gross profitability, ROA, and momentum and is updated monthly, as in Stambaugh and Yuan (2017).

- *CMA*: Investment is measured by asset growth. It is updated annually in June, as in Fama and French (2015).
- *MGMT_m*: Management is a composite factor comprising asset growth, composite equity issuance, investment-to-assets, net operating assets, and operating accruals and is updated monthly, as in Stambaugh and Yuan (2017).
- *WML_m*: Winner and loser stocks are determined by past 12 month returns, excluding the most recent month. It is updated monthly, as in Carhart (1997).
- *PMO_m*: Pessimistic and optimistic sentiment is measured by abnormal turnover and is updated monthly, as in Liu, Stambaugh, and Yuan (2019).

Table 1 contains descriptive statistics of the individual factor return series for the Chinese A-share market over the period 2000 to 2019. The equity risk premium has been positive and economically close to what we have seen over longer periods in developed markets with 0.57% per month but is statistically indistinguishable from zero. This is partially due to the relatively high stock market volatility in combination with a short sample period.

[INSERT TABLE 1 HERE]

Average factor returns are similar to the returns documented in the recent literature. In the text, we describe the raw factor returns, but the table shows that CAPM alphas are economically close to the raw returns and statistical inference on them leads to the same conclusion. The average return of 0.39% for the size factor is slightly less than half of the 1.01% and 1.03% that Lin (2017) and Liu, Stambaugh, and Yuan (2019) report over their sample period from 1997 to 2015 and 2000 to 2016, respectively. When we form portfolios similarly and restrict our sample to end in 2016, we can replicate their result.¹³ The size factor has seen substantially negative returns for the period 2017 to 2019.¹⁴

¹³ Liu, Stambaugh, and Yuan (2019) orthogonalize their size factor only to (earnings-based) value and growth, while we orthogonalize it to each of our 11 factors.

¹⁴ Chen and Chi (2018) find that Chinese mutual funds create alpha by successfully timing the size factor.

The value factor has positive returns with a similar Sharpe ratio as the SMB factor. Updating this factor on a monthly basis doubles returns for similar volatility and using the earnings- instead of the book-based value factor increases returns further to 1.17% per month, similar to the 1.27% per month that Liu, Stambaugh, and Yuan (2019) report for the period 2000 to 2016.

For profitability, we also see that it matters how this is measured. The operating profitability measure, as defined in Fama and French (2015), yields a factor return close to zero, which only slightly improves when using a cash-based version. However, the monthly updated ROE_m and the composite PERF_m factors have economically and statistically significant excess returns of 0.80% and 0.62% per month, respectively. The finding that a profitability factor based on ROE performs better than more top-line profitability factors is in line with Guo et al. (2017), who find that for the Chinese stock market, ROE has higher predictive power than profitability measures based on gross profits or free cash flow.

The investment factor is known to be weak in the Chinese A-share market; see, e.g., Guo et al. (2017), Lin (2017), and Huang (2019). We confirm this for both the CMA factor from the Fama and French (2015) model and the MGMT_m factor from the Stambaugh and Yuan (2017) model.

The two sentiment variables are clearly measuring different phenomena. Return momentum effects (WML_m) are absent from the A-share market, while abnormal trading volume (PMO_m) has a statistically significant 0.88% per month excess return. The weak performance of return momentum is consistent with Hsu et al. (2018), Lin (2019), and Gao, Han, Li, and Xiong (2021), and the significant excess returns of PMO_m have been documented by Liu, Stambaugh, and Yuan (2019).

Governance of China A-share companies may be different than for many other markets, as a relatively large part of the market consists of State Owned Enterprises (SOEs), where the Chinese government has considerable influence over business decisions. Liao et al. (2014) study the privatization effect in China by comparing the operational performance (such as revenue, profitability, and employment) and stock performance of SOEs before and after the Split-Share Structure Reform. They report significant increases in both types of performance after privatization. Their evidence points to the existence of additional agency problems in SOEs, which give rise to a conflict of interest between government agents and public investors. These agency problems may induce governance problems, which is subsequently reflected in a firm's operational and financial

performance. However, Jansen, Swinkels, and Zhou (2021) report factor returns for a subsample that excludes SOEs, and a subsample that includes only SOEs. The differences between the factor returns on both non-overlapping samples are very small, on average, the factor return is 0.79% per month for SOEs and 0.76% per month for non-SOEs. Moreover, in many cases, the same factors are significant or insignificant in both subsamples. Their empirical evidence suggests that the distinction between SOEs and non-SOEs is not so important when evaluating factor performances.

They also report that excess returns of factors are roughly half caused by the long side of the factors, and half by the short side of the factors. Hence, while short-sale constraints may be more relevant in the China A-share market than in other markets, they are not driving the observed factor returns.

The final column of Table 1 reports the average monthly factor turnover. Factor turnover is calculated as the sum of the one-way turnover of the long and short legs, which both average the big and small legs. This means that the maximum monthly turnover is 200%. Turnover is lowest for yearly rebalanced factors such as HML, RMW_OP, and CMA at 8.27%, 7.52%, and 11.56%, respectively. The monthly updated version of HML, HML_m, has a turnover of 33.70%, while HML_EP_m has a turnover of 44.81%. Since valuation variables are relatively persistent, increasing the updating frequency by 12 (going from yearly to monthly rebalancing) results in 'only' 4 to 5 times higher turnover. Similar patterns are observed within the profitability category. Unsurprisingly, the two sentiment variables have very high turnover, 64.62% and 106.46% for WML_m and PMO_m, respectively. Accounting for transaction costs would likely push WML_m into negative territory, while the average realized return on the PMO factor would also be substantially lower in a realistic portfolio management setting, even after including transaction cost mitigation techniques described in Novy-Marx and Velikov (2019).

Table 2 displays the correlation matrix of the factor returns over our sample period. We can thus gauge whether the individual factors belonging to value, profitability, investment, and sentiment groups are in fact similar. The correlation matrix also informs about which factor definitions diversify well with respect to others, which is important for obtaining low pricing errors in a multi-factor asset pricing model.

[INSERT TABLE 2 HERE]

Table 2 shows that the factors within value, profitability, and investment factor groups behave similarly. The correlations between the value factors range from 0.33 to 0.87, between the profitability factors from 0.39 to 0.75, and between CMA and MGMT_m the correlation is 0.63. However, the sentiment group contains two variables that are virtually uncorrelated (0.06). Although the grouping is inconsequential for our further analyses, it is important to realize that correlation between these two is low since both measures are often associated with sentiment. The correlation matrix also shows that the correlation between earnings-based value and profitability factors is rather high, with a correlation of 0.79 between HML_EP_m and RMW_ROE_m. This high correlation is expected as the two factors share the same numerator. In contrast, the correlation between the book-based value factors and profitability factors is much lower, even close to zero. The correlation between momentum and the value factors based on the book-to-market ratio is -0.57, while the correlation with the earnings-based value factor is only 0.09. These diverging correlations between factor pairs already indicate varying diversification gains when combining factors in factor models.

3. Comparing asset pricing models in the Chinese A-share market

3.1. Factor models

The model selection follows Barillas et al. (2020) and Hanauer (2020). We include eight common factor models that have been developed for U.S. equity markets. Additionally, we include two factor models that we tailor to price the cross-section of the Chinese A-share market. Table 3 provides an overview of the different factor models.

[INSERT TABLE 3 HERE]

The Fama and French (1993) three-factor model is the most popular multi-factor model used to price the cross-section of U.S. equity returns. In addition to the market factor from the Capital Asset Pricing Model, it contains a size and value factor. Motivated by a rewritten dividend discount model, Fama and French (2015) extend the set of factors to five by adding an investment and a profitability factor. Fama and French (2018) further extend the model with a sixth factor,

momentum. We also consider slight variations of the six-factor model. One variation is to use a monthly instead of an annually updated value factor and a cash-based operating profitability factor (cf., Fama and French, 2018; Barillas et al., 2020). Another variation is to replace the profitability factor with the ROE profitability factor from the q-factor model from Hou, Xue, and Zhang (2015). We also consider the q-factor model itself, which is built on the neoclassical q-theory of investment and the latter model minus value and momentum, leaving four pricing factors.

Furthermore, we include the Stambaugh and Yuan (2017) mispricing model rooted in behavioral finance. The model contains four factors: market and size factors common to the other models and two mispricing factors. These two mispricing factors are performance and management, a combination of three and five indicators, as described in Section 2.2. These two factors can be loosely interpreted as profitability and investment factors.

The two models that are specifically developed for the Chinese A-share market are taken from Liu, Stambaugh, and Yuan (2019), one with three factors and the other with four. The three-factor model contains the market, size, and earnings-based value factor and is, therefore, a modified version of the Fama and French (1993) three-factor model. As we have seen, the earnings-based value factor has higher excess returns than the conventional book equity-based value factor. The four-factor model adds a sentiment factor that the authors label Pessimistic Minus Optimistic (PMO) and is derived from abnormal trading volume. This fourth factor is motivated by the specific structure of the Chinese market. The majority of share trading is done by retail investors with a short investment horizon. The theoretical motivation is from Baker and Stein (2004) and Baker and Wurgler (2006), but this factor has not been extensively documented as an important factor in other markets.

3.2. Methodology

Following Barillas et al. (2020) and Hanauer (2020), we compare the asset pricing models by ranking them based on their maximum squared Sharpe ratios. Thereby, the statistical inference is conducted based on the asymptotic analysis derived by Barillas et al. (2020). The maximum Sharpe ratio model comparison follows from Barillas and Shanken (2017), who show that test assets are irrelevant when conducting model comparison tests. In other words, the asset pricing model with the highest squared Sharpe ratio is equivalent to the model with the least mispricing of the test

assets. Nevertheless, we show how well the models can price different test assets in Tables A4 and A5 in the Appendix.¹⁵

The model comparison consists of three steps. First, maximum sample squared Sharpe ratios are calculated for each model as $\mu'\Sigma^{-1}\mu$, where μ is the vector with average factor returns and Σ is the covariance matrix of factor returns over the sample period. We adjust the maximum squared Sharpe ratio for each model to be unbiased in small samples under joint normality by multiplying it by $(T - K - 2)/T$ and subtracting K/T , where T is the number of return observations and K is the number of factors. Second, we calculate the differences between the bias-adjusted maximum sample squared Sharpe ratios for all pairs of factor models. Third, we compute p-values for the test of equality of the maximum squared Sharpe ratios as in Barillas et al. (2020). For this computation, we have to differentiate between nested and non-nested models. In nested models, all smaller model factors are contained in the other model, while in non-nested models, both models contain factors not included in the other model.

For nested models, we test whether the model's maximum squared Sharpe ratio with more factors is higher than the squared Sharpe ratio of the model with fewer factors. This is equivalent to testing whether the factors in the larger model that are not also in the smaller model produce significant alphas when regressed on the smaller model. For example, because the FF3 model is nested in the FF5, we test whether the FF3 alphas of RMW_OP and CMA are zero. The statistical test for multiple excluded factors, as in this example, is the application of the (GRS) test in Gibbons, Ross, and Shanken (1989). For a single excluded factor, the corresponding test is whether the intercept in the regression of the excluded factor on the nested model is significantly different from zero.

When the models are non-nested, we conduct a sequential test. First, we determine whether the maximum squared Sharpe ratio of the model that includes all of the factors from both models is larger than for the model composed of only the common factors.¹⁶ This is equivalent to a test for

¹⁵ While test assets may contain any asset, we focus our attention on the joint set analyzed in Hsu et al. (2018), Liu, Stambaugh, and Yuan (2019), and this study. This gives a set of 30 anomalies, 18 of which are not included in one of the investigated factor models. The definition of these anomalies is outlined in Appendix C, and their performance can be found in Table A3 in the Appendix. The results presented in Table A4 demonstrate that the model ranking depends crucially on the choice of test assets. When all 30 anomalies are included as test assets, the order of the model ranking indeed mirrors the model ranking of the maximum Sharpe ratio test. However, when we exclude the RHS factors from the set of test assets, the order of the model changes. By removing strong factors from the set of test assets, the GRS statistics for all models decreases, but more for the models that had problems with pricing the factors of other models. Table A5 contains results with 10 industry portfolios as test assets. The results from this set of test assets show that the China-specific asset pricing models perform best.

¹⁶ As all investigated models include the market and size factors, the models have at least two common (overlapping) factors.

nested models as described above. For example, because the FF3 and LSY3 models are non-nested, we conduct the GRS test of the non-common factors, HML and HML_EP_m, on the common factors model (RMRF and SMB). If the null hypothesis cannot be rejected, then the evidence is consistent with the notion that the squared Sharpe ratio of the model with all factors equals that of the model consisting only of the common factors. This means that the common factors model is as good as the models that add the non-common factors. Thus, the non-common factors are redundant, and the two non-nested models are equivalent under the null. If this preliminary test is rejected, some or all of the non-common factors are not redundant and contribute to an increase in the squared Sharpe ratio compared to the common factors model. However, the preliminary test does not tell us which non-nested model has a higher squared Sharpe ratio, and equality may or may not hold for the two models. In this case, we conduct the direct test of whether the squared Sharpe ratios are different, as described in Barillas et al. (2020). The associated test statistic is calculated as the (bias-adjusted) squared Sharpe ratio difference divided by its standard error. The standard error of the squared Sharpe ratio difference is the square root of the asymptotic variance divided by the number of monthly observations; see also Proposition 1 in Barillas et al. (2020).

The pairwise model comparison may overstate statistical significance when comparing many competing models. Therefore, we also consider the multiple-model comparison by Barillas et al. (2020). Given a set of competing models, this joint test determines whether the benchmark model is at least as good as the other models, i.e., has the highest squared Sharpe ratio. The null hypothesis is that none of the alternative models has a higher squared Sharpe ratio than the benchmark model. As with the pairwise comparison, a distinction is made between nested and non-nested models. For the former, a single expanded model is constructed containing all of the factors from the alternative models. We can then use the pairwise test on the expanded and benchmark models. Comparison of non-nested models is more complex and relies on the multivariate inequality analysis of Wolak (1989). A likelihood-ratio test statistic can be computed to test the null, where a larger test statistic implies that the null hypothesis does not hold, i.e., that some other model has a higher squared Sharpe ratio than the benchmark.

3.3. Comparing asset pricing models

Table 3 contains an overview of the factors used in each asset pricing model, their maximum squared monthly Sharpe ratios, the p-value for the null hypothesis that the model's maximum

squared Sharpe ratios are equal to zero, and the factors' weight in the ex-post maximum squared Sharpe ratio portfolios.¹⁷

On the left-hand side of Table 3, we observe that the Fama and French (1993, 2015, 2018) three, five, and six-factor models have Sharpe ratios that are statistically indistinguishable from zero at the 5% significance level for the Chinese A-share market. Adding both the profitability (RMW_OP) and investment (CMA) factors does not increase the Sharpe ratio with respect to the three-factor model. This is partially due to the bias adjustment.¹⁸ Table 4, which contains the differences in maximum squared Sharpe ratios and associated p-values, shows that the models are equivalent (p-value 0.32) and 'standard' profitability and investment have little to no added value in China. The Sharpe ratio only becomes statistically significantly different from zero when a cash-based alternative replaces the standard profitability factor. The Sharpe ratio is further increased when a monthly updated value factor is used. This model emerged as the best-performing asset pricing model for international markets in Hanauer (2020). The q-factor model by Hou, Xue, and Zhang (2015) performs narrowly better with a monthly Sharpe ratio of 0.447, as it is a parsimonious model with only four factors and developed for the U.S. equity market. However, the difference is not statistically significant (p-value 0.99). One interesting observation emerges when looking at the factor weights of the maximum squared Sharpe ratio portfolios: models comprising both a value (HML or HML_m) and an investment factor (CMA) have positive weights for the value factor but negative weights for the investment factor. Therefore, the evidence from Fama and French (2015) that value is redundant when controlling for investment does not carry over to the China A-share market.

[INSERT TABLE 4 HERE]

The Stambaugh and Yuan (2017) mispricing model also has four factors and was developed for the U.S. equity market but has a significantly (p-value 0.03) lower maximum Sharpe ratio than the

¹⁷ The test of the null hypothesis that the model's maximum squared Sharpe ratios are equal to zero is evaluated using a χ^2 test of $T \cdot SR^2 \sim \chi_K^2$, where SR^2 is the estimated squared Sharpe ratio of the factor model.

¹⁸ Without the bias adjustment, the Sharpe ratios are 0.164, 0.192, and 0.224 for the Fama-French three-, five-, and six-factor models, respectively.

q-model, as shown in Table 4. Conversely, the three-factor model by Liu, Stambaugh, and Yuan (2019) performs significantly better than the two four-factor models with a monthly Sharpe ratio of 0.570. We have to keep in mind that this is due to its calibration on the Chinese A-share market, for which the earnings-based value factor has double the outperformance compared to the book-based value factor. The four-factor model by Liu, Stambaugh, and Yuan (2019) performs even better, thanks to the use of the rather ‘exotic’ sentiment factor that works particularly well for the Chinese A-share market. Note, however, that this model is not statistically better than the six-factor model where the profitability factor is replaced with ROE_m (FF6_ROE), i.e., the profitability factor from the q-factor model.

The multiple-model comparison tests of the squared Sharpe ratios presented in the bottom panel of Table 4 reveal similar results. The null hypothesis that “none of the alternative models is superior” can be rejected for most of the models: the Fama and French (1993, 2015, 2018) three-, five-, and six-factor models, the two modified Fama-French six-factor models with cash-based operating profitability factors, as well as Hou, Xue, and Zhang (2015) and Stambaugh and Yuan (2017) four-factor models. Only for the Liu, Stambaugh, and Yuan (2019) three- and four-factors models and the modified Fama-French six-factor model with a monthly updated value and a profitability factor based on ROE, we cannot reject that none of the alternative models is superior.

The results presented so far are based on the full sample period from July 2000 to December 2019. As Barillas and Shanken (2018), Guidolin, Hansen, and Lozano-Banda (2018), and Kan, Wang, and Zheng (2022) highlight, there is likely to be an upward bias in the sample Sharpe ratios of the models identified as the best performers. Therefore, it is interesting to also investigate the out-of-sample performance of the models. Following Hanauer (2020), we apply an estimation period for determining tangency portfolio weights for the different models. These factor weights are then applied to calculate portfolio returns in out-of-sample performance evaluations. Table 5 presents the squared out-of-sample Sharpe ratios for the different models for rolling estimation periods of 30, 60, and 120 months.

[INSERT TABLE 5 HERE]

The results in Table 5 show that the Liu, Stambaugh, and Yuan (2019) four-factor model achieves the highest squared out-of-sample Sharpe ratio for three of the four rolling estimation windows. Only for 120-month estimation period, the Liu, Stambaugh, and Yuan (2019) three-factor model is slightly better. As expected, the squared out-of-sample Sharpe ratios are lower than their in-sample counterparts. However, the ranking of the models remains nearly unchanged, and the squared out-of-sample Sharpe ratios of the Liu, Stambaugh, and Yuan (2019) four-factor model are all above 0.25, and therefore about two-thirds of the maximum squared Sharpe ratio of the full in-sample period.

Summarizing Tables 3 to 5, we can conclude that China-specific asset pricing models are better than off-the-shelf models developed for the U.S. However, whether this is because they inherently are better at capturing the true return drivers of the Chinese equity market or because they are the result of data mining remains an open question. From the asset pricing models originally developed for the U.S. equity market, the q-factor model and the Fama and French (2018) six-factor model with a cash-based operating profitability factor and a monthly updated HML factor are preferred.

4. A data-driven asset pricing model

In previous sections, we compared common asset pricing models within a frequentist testing framework. However, Shanken (1987) highlights the advantages of using Bayesian approaches over frequentist models for asset pricing tests.¹⁹ For example, Pastor (2000), Pastor and Stambaugh (2000), Carrasco and Hansen (2022), and Hansen (2022) use Bayesian methods for optimal portfolio choice when investors are to a varying degree skeptical about asset pricing models and therefore also want to incorporate historical returns data. These papers on portfolio selection show that combining prior information from asset pricing models with realized returns data leads to more effective portfolios than with a focus on either models or data. Our goal is to find the best asset pricing model for the Chinese A-share market given prior information on asset pricing models that have been successful in the US equity market and combine this with realized data on the Chinese A-share market. In this section, we extend upon this and apply a Bayesian data-driven

¹⁹ For example, frequentist p-values indicate the probability that under the null hypothesis the observed event or a more extreme one occurs. While such p-value may be useful for some applications, a direct comparison between the likelihood of the occurrence of an event under the null hypothesis and the alternative hypothesis would be more suitable for other applications.

approach to derive the best asset pricing model for the Chinese A-share market. Specifically, we make use of the Barillas and Shanken (2018) and Chib, Zeng, and Zhao (2020) Bayesian asset pricing framework in which candidate factor models are simultaneously tested against each other. More specifically, given a set of K candidate factors, we form all 2^K possible combinations of the factors and derive the posterior support for each model. This way, we employ a ‘kitchen sink’ approach in which the data determines the preferred factor model after specifying the input factors.

We again use the key result from Barillas and Shanken (2017) that, for the purpose of model comparison, the choice of test assets is irrelevant and only a factor model’s ability to price the excluded factors (i.e., the factors not part of the current model) matters. As such, inference on model comparison is derived from the aggregation of regressions of excluded factors on included factors, corresponding to all possible combinations of factor models. Based on these regressions, marginal likelihoods are calculated for each model. Marginal likelihoods quantify the comparative support in the data for a model. To promote parsimony, marginal likelihoods contain a component that penalizes the support for models with redundant factors.

[INSERT FIGURE 1 HERE]

Barillas and Shanken (2018), henceforth BS, derive closed-form expressions for the marginal likelihoods. However, Chib, Zeng, and Zhao (2020), henceforth CZZ, show that these likelihoods are theoretically unsound. BS specify separate improper priors for each of the nuisance parameters. This renders marginal likelihood-based model comparison invalid, as the marginal likelihoods are indeterminate, i.e., they are only identified up to an arbitrary positive constant. Using a change-of-variable property, CZZ derive a new class of improper priors that shares a common improper prior across all models. This ensures that the arbitrary constant is the same across marginal likelihoods, validating model comparison. We implement the BS model comparison framework using the correct CZZ marginal likelihoods on the components of the Fama-French (2018) six-factor model with cash-based operating profits, the Hou, Xue, and Zhang (2015) q-factor model, and the Liu, Stambaugh, and Yuan (2019) four-factor model for the China A-share market. Since our sample is only 234 months, the BS methodology cannot accommodate more factors. In their simulation

study, CZZ require a sample size of 12,000 months for the true data generating process to be selected in about 80% of the cases, when 12 factors are included. For their sample size of 600 months, which is about double that of ours, the hit rate decreases to 20%. This is not surprising since their model space is much larger than the time series length. Hence, we need to rely on some prior information to limit the cardinality of the model space; otherwise, the model probabilities would hardly deviate from their near-zero prior.

Figure 1 contains the recursive model probabilities, which show how posterior beliefs evolve as more data becomes available, from the CZZ methodology with uniform prior model probabilities and a Sharpe ratio multiple of 3.²⁰ The data-driven preferred model for Chinese A-shares contains (in addition to RMRF) SMB, HML_m, HML_EP_m, RMW_ROE_m, WML_m, and PMO_m. This seven-factor model is preferred over a model that additionally includes CMA or RMW_CbOP. Furthermore, this model (denoted as CZZ) is also superior to all other models under the classical maximum squared Sharpe ratios metric, as shown in Table 4. However, we cannot reject that none of the alternative models is superior in the multiple-model test.

The model with the highest likelihood contains both sentiment variables, WML_m and PMO_m. This is somewhat surprising, as momentum comes with a low factor premium in China. However, the correlation between the sentiment measures is low and momentum also offers large diversification benefits with the other factors. Notably, as seen in Table 2, momentum has a strong negative correlation with HML_m of -0.57, which allows it to add value in a multi-factor model despite its small premium. The negative correlation between value and momentum is not unique to the Chinese A-shares market but generally observed across markets; see Asness, Moskowitz, and Pedersen (2013). Our results suggest that momentum is not redundant when paired with a book-based value measure. A similar result is obtained by Asness (2011) for the Japanese equity market, where momentum is also a weak standalone factor.

[INSERT FIGURE 2 HERE]

²⁰ We use uniform prior model probabilities to avoid favoring certain models over others. The Sharpe ratio multiple of 3 follows from Chib, Zeng, and Zhao (2020).

Figure 2 contains the cumulative factor probabilities. Cumulative factor probabilities are the sum of the model probabilities for all models that include that factor. These probabilities give the relative importance of each factor. Since the market factor is judged to be part of each model from a theoretical basis, it is not included in the figure. It is clear that the size factor is very important, converging to 100% after 2015. The sentiment factor PMO_m is the second most important factor with nearly 90% probability. Although the cumulative probabilities of both value factors are similar with 80% at the end of the sample, there have been periods in which HML_m had a substantially higher probability (2008 to 2012) and a substantially lower probability (2015 to 2018). The importance that value has shown in pricing Chinese A-shares is an important contribution to the asset pricing literature, as recent literature on the U.S. casts doubt on the usefulness of this factor; see, e.g., Lev and Srivastava (2022). The next two factors (RMW_ROE_m and WML_m) are substantially below with about 70% probability. The final two factors have a cumulative probability slightly below the starting point of 50%. These two factors do not appear in the optimal data-driven asset pricing model.

Remarkably, HML_m, HML_EP_m, and RMW_ROE_m are jointly part of the model with the highest likelihood, given that the underlying signals are strongly interrelated, i.e., that E/P equals ROE times B/M. Is this an outcome of the Bayesian asset pricing framework or are neither of the factors redundant using classical statistics? Table 6 answers this question with spanning regressions, i.e., we iteratively regress each of the non-market factors' returns on the other factors of the seven-factor model.

[INSERT TABLE 6 HERE]

We find that the HML_m factor is statistically significant with a t-value of 3.28, while HML_EP_m and RMW_ROE_m are only significant at the 10% level with t-statistics of 1.75 and 1.74, respectively. These numbers show that HML_m has the highest marginal contribution while HML_EP_m and RMW_ROE_m are somewhat less important when controlling for other factors, although not redundant. These results are consistent with Liu, Shi, and Lian (2019), who also find that multiple value factors are required to price the cross-section of stock returns in Chinese A-

shares. The ordering of value factors is different from the standalone analysis in Table 1, where the former had a t-statistic of 2.58 and the latter t-statistics of 5.60 and 3.84. This observation reinforces the notion that interaction effects and correlations between factors are important for choosing asset pricing models.

Furthermore, we find that the size factor SMB and the sentiment factor PMO_m are statistically significant with t-statistics of 5.90 and 3.14, respectively, and the sentiment factor WML_m is borderline significant with a t-statistic of 2.02. The latter is substantially larger than the t-statistic of 0.59 for the momentum premium in isolation, as displayed in Table 1.

In summary, we select the nine most promising factors from existing asset pricing models and a data-driven approach suggests that seven of these are important for pricing Chinese A-shares.

5. Comparing asset pricing models with transaction costs

Our analysis thus far follows the majority of studies by ignoring transaction costs when calculating factor returns and conducting model comparison tests. However, Detzel, Novy-Marx, and Velikov (2023) show that the ranking of competing asset pricing models may substantially change when considering transaction costs. The underlying reason is that ignoring transaction costs favors models that include more frequently updated factors with less persistent characteristics. However, Novy-Marx and Velikov (2016) show that such high turnover factors often do not generate significant net returns despite their high gross returns. Therefore, the maximum squared-Sharpe ratios reported in the previous sections do not come close to spanning the actual achievable mean-variance efficient frontier for this set of factors. The achievable frontier more closely represents the investment opportunity set attainable for investors. To better account for such frictions, we switch from the maximum gross squared Sharpe ratio to the maximum net (achievable) squared Sharpe ratio metric, which is calculated using net of transaction costs factor returns.²¹

To calculate net factor returns, we deduct estimated transaction costs from the gross factor returns presented in Section 2. We estimate these monthly transaction costs by multiplying the monthly

²¹ We admit that we only investigate the achievable frontier by combining separate factors. The achievable Sharpe ratio might be higher when factor scores are first combined before forming a composite portfolio. For more details on this 'mixed vs. integrated debate,' we refer to Bender and Wang (2016), Clarke et al. (2016), and Leippold and Rueegg (2018).

factor turnover with a transaction cost estimate of 74 basis points per round-trip transaction, i.e., 37 basis points per single-trip transaction. The 37 basis points estimate comes from Guo, Zhao, and Wang (2021), who examine the quoted and effective spread of 1,265 stocks listed on the Shanghai and Shenzhen stock exchange over the period 2009-2016. They find that the spreads vary between 11 and 37 basis points for their sample. To be conservative, we opt for the higher range of their estimate as transaction costs also include commissions and stamp tax.²² In addition, Leippold, Wang, and Zhou (2022) estimate 25 basis points to be reasonable trading costs under normal circumstances but acknowledge that costs can be higher during market turmoil. This estimate seems to be consistent with our estimate of 37 basis points. With the resulting net factor returns, we can now recompute maximum net squared Sharpe ratios for different factor models and compare them across models.

We calculate model turnover by multiplying the in-sample tangency portfolio weights for each asset pricing model and the average factor turnover.²³ Turnover is reported in the bottom row of Table 3. The turnover analysis reveals some interesting findings. First, the turnover of the FF3 model is only 12.65%, about three times lower than the 35.67% of its Chinese counterpart LSY3. However, FF3's maximum Sharpe ratio of 0.117 is nearly five times lower than LSY3's Sharpe ratio of 0.570. Therefore, we expect that up to a certain level of transaction costs, the LSY3 model would perform better than the FF3 model in the Chinese A-share market. Second, turnover drops from 19.93% to 13.85% between FF6 and FF6_CP. This can be explained by the FF6 tangency portfolio allocating 32% to WML (facilitated by a large short position in CMA), driving up the turnover. Third, the difference in the maximum Sharpe ratio between FF6_CPM and Q4 is not statistically significant, but the turnover of FF6_CPM is 6%-points lower. Lastly, PMO_m inflates the turnover of the LSY4 model to 47%.

We acknowledge that our approach of including fixed trading costs to account for turnover in these periodically sorted portfolios has limitations. Stock- and time-specific transaction cost estimates for specific trade sizes would allow us to derive more precise cost estimates for realistic portfolio

²² The stamp tax on both the Shanghai and Shenzhen stock exchanges was 40 bps on both the buy and the sell side in 2000 but reduced to 20 bps in November 2001 and to 10 bps in January 2005. In May 2007, the tax was increased to 30 bps but reverted back to 10 bps in April 2008. Since September 2008, the stamp tax is only charged on the sell side (cf., Hu et al., 2021). According to Leippold et al. (2022), commission fees are typical smaller than stamp tax and were around 5 bps in 2012 for institutional investors and below 2 bps in recent years.

²³ We assume that turnover in the market portfolio is zero.

management applications. However, since the required data for estimating precise transaction costs is not available to us, this approach is infeasible. Moreover, smarter portfolio construction rules as outlined in Novy-Marx and Velikov (2019), can further mitigate transaction costs. Because there are so many degrees of freedom in designing smarter portfolio construction rules to reduce trading costs, such an approach warrants a separate study. Nevertheless, the way we deal with transaction costs is commonly applied in the literature (cf., De Roon, Nijman, and Werker, 2001, De Groot, Pang, and Swinkels, 2012; Wang and Yan, 2021).

In Table 7, we present the outcome of the model comparison after including trading costs. Panel A shows the gross and net maximum squared Sharpe ratios and the associated model turnover figures. The gross values are the same as in Table 3, but we repeat them for a better comparison. We can make two general observations. First, taking transaction costs into account leads to much lower maximum net squared Sharpe ratios for all models. The average maximum gross squared Sharpe ratio of 0.193 across all models drops to an average maximum net Sharpe ratio of 0.063, a decrease of 67%. However, all models with statistically significant maximum gross Sharpe ratios also have statistically significant maximum net squared Sharpe ratios, except for the Stambaugh and Yuan (2017) mispricing model.²⁴ Second, the average model turnover of 30% drops to 27%, showing that the net of transaction costs analysis indeed favors lower turnover factors.

[INSERT TABLE 7 HERE]

We observe some important differences in the ordering of models based on the net instead of the gross squared Sharpe ratio.²⁵ Whereas the Q4 model's gross ratio was economically and statistically significantly higher than that of the FF3, FF5, and FF6 models, this is no longer the case when transaction costs are included. This is due to the turnover of 30.63% for the Q4 model, while the turnover of the FF models is less than 20%.

²⁴ The turnover of this model increases from 50.43% to 53.09%, because the weight in the management factor slightly increases.

²⁵ Results are similar for the multiple-model comparison test. The performance of models with high-turnover factors deteriorates, while we can no longer reject the null hypothesis that none of the other models is superior to the benchmark for both the yearly updated FF6_CP and the monthly updated FF6_CPm models.

The LSY4 model previously had the highest ratio among the existing asset pricing models of 0.390. However, given its high turnover of 47.00%, its net ratio reduces more than that of the LSY3 model, which only had a 35.93% turnover. When we include trading costs, the squared Sharpe ratio of the LSY4 model decreases to 0.142, as does the turnover. This is because PMO_m, which had the highest turnover in Table 1, now receives a substantially lower weight. The LSY3 model has a similar turnover after including transaction costs, and its net ratio is 0.145. While Table 4 showed that the LSY4 model was significantly better than the LSY3 model, the difference is no longer statistically significant (p -value = 0.57) after including transaction costs.

The data-driven CZZ model, which had the highest squared Sharpe ratio of 0.431, now drops to third place with a ratio of only 0.131. Although not significantly different from the LSY3 and LSY4 models, we again see that high-turnover factor models are punished and lose their edge compared to models with lower turnover. Figure A1 in Appendix A shows that when we repeat the data-driven asset pricing procedure to find the ‘CZZ net’ model, the top model is now equal to the LSY3 model, which was also found to have the highest Sharpe ratio in the pairwise comparison. Figure A2 demonstrates that SMB and HML_EP_m indeed have the highest factor probabilities of more than 90% after accounting for transaction costs. In contrast, the probability of the remaining factors barely differs from the prior probability of 50%.

In short, our empirical results resonate with those of Detzel, Novy-Marx, and Velikov (2023) found for the U.S. stock market. Incorporating transaction costs when constructing factor models is important as it materially changes the ranking of competing asset pricing models by penalizing models with high-turnover factors that cannot be exploited in a realistic portfolio management setting. The preferred model based on both the direct model comparison and data-driven approach is a three-factor model consisting of the market, size, and the monthly updated earnings-based value factor.

6. Conclusion

We compare the performance of various asset pricing models for the Chinese A-share equity market. We find that the conventional three- and five-factor models by Fama and French (1993, 2015) poorly price assets in the Chinese A-share market. The q-factor model and a modified Fama-

French six-factor model perform relatively well, but an asset pricing model tailored to the Chinese market by Liu, Stambaugh, and Yuan (2019) performs even better. This model contains an earnings-based instead of a book-based value factor and introduces a ‘pessimistic minus optimistic’ sentiment factor based on abnormal trading volume. A data-driven asset pricing framework extends this China-specific asset pricing model with three additional factors.

When we take transaction costs into account, the ranking of models changes as high-turnover factors, such as the sentiment factors, experience larger drops in factor premiums than lower-turnover factors. Furthermore, the preferred data-driven asset pricing model, when accounting for transaction costs turns out to be the Liu, Stambaugh, and Yuan (2019) three-factor model with the market, the size, and the monthly updated earnings-to-price value factor. In contrast to Liu, Stambaugh, and Yuan (2019), who find that a four-factor model best prices the Chinese cross-section of stock returns, we find that their three-factor model is best.

References

- Arnott, R., Harvey, C., Kalesnik, V., and Linnainmaa, J. (2021) “Reports of value’s death may be greatly exaggerated”, [*Financial Analysts Journal*](#) 77(1), pp. 44-67.
- Asness, C. (2011) “Momentum in Japan: The exception that proves the rule”, [*Journal of Portfolio Management*](#) 37(4), pp. 67-75.
- Asness, C., and Frazzini, A. (2013). “The devil in HML’s details”, [*Journal of Portfolio Management*](#) 39(4), pp. 49-68.
- Asness, C., Moskowitz, T., and Pedersen, L. (2013) “Value and momentum everywhere”, [*Journal of Finance*](#) 68(3), pp. 929-985.
- Baker, M., and Stein, J. (2004) “Market liquidity as a sentiment indicator”, [*Journal of Financial Markets*](#) 7(3), pp. 271-299.
- Baker, M., and Wurgler, J. (2006) “Investor sentiment and the cross-section of stock returns”, [*Journal of Finance*](#) 61(4), pp. 1645-1680.
- Ball, R., Gerakos, J., Linnainmaa, J., and Nikolaev, V. (2016) “Accruals, cash flows, and operating profitability in the cross section of stock returns”, [*Journal of Financial Economics*](#) 121(1), pp. 28-45.
- Barillas, F., Kan, R., Robotti, C., and Shanken, J. (2020) “Model comparison with Sharpe ratios”, [*Journal of Financial and Quantitative Analysis*](#) 55(6), pp. 1840-1874.
- Barillas, F., and Shanken, J. (2017) “Which alpha?”, [*Review of Financial Studies*](#) 30(4), pp 1316-1338.
- Barillas, F., and Shanken, J. (2018) “Comparing asset pricing models”, [*Journal of Finance*](#) 73(2), pp. 715-754.
- Bender, J., and Wang, T. (2016) “Can the whole be more than the sum of the parts? Bottom-up versus top-down multifactor portfolio construction”, [*Journal of Portfolio Management*](#) 42(5), pp. 39-50.
- Blitz, D., and Hanauer, M. (2021) “Resurrecting the value premium”, [*Journal of Portfolio Management*](#) 47(2), pp. 63-81.
- Blitz, D., Hanauer, M., and van Vliet (2021) “The volatility effect in China”, [*Journal of Asset Management*](#) 22(5), pp. 338-349.
- Cakici, N., and Zaremba, A. (2022) ”Salience theory and the cross-section of stock returns: International and further evidence”, [*Journal of Financial Economics*](#) 146(2), pp. 689-725.
- Cakici, N., Fieberg, C., Metko, D., and Zaremba, A. (2023) “Do Anomalies Really Predict Market Returns? New Data and New Evidence”, [*Review of Finance*](#), forthcoming.
- Cosemans, M., and Frehen, R. (2021) “Salience theory and stock prices: Empirical evidence”, [*Journal of Financial Economics*](#) 140(2), pp. 460-483.
- Carpenter, J., Lu, F., and Whitelaw, R. (2021) “The real value of China’s stock market”, [*Journal of Financial Economics*](#) 139(3), pp. 679-696

- Carhart, M. (1997) “On persistence in mutual fund performance”, [*Journal of Finance* 52\(1\)](#), pp. 57-82.
- Carrasco, I., and Hansen, E. (2022) “Asset pricing model uncertainty and portfolio choice”, [*Finance Research Letters* 45](#), 102144.
- Cheema, M. A., Nartea, G. V., & Man, Y. (2020). “Maxing out in China: Optimism or attention?”, [*International Review of Finance* 20\(4\)](#), pp. 961-971.
- Chen, Q., and Chi, Y. (2018) “Smart beta, smart money”, [*Journal of Empirical Finance* 49](#), pp. 19-38.
- Chib, S., Zeng, X., and Zhao, L. (2020) “On comparing asset pricing models”, [*Journal of Finance* 75\(1\)](#), pp. 551-577.
- Clarke, R., De Silva, H., and Thorley, S. (2016) “Fundamentals of efficient factor investing”, [*Financial Analysts Journal* 72\(6\)](#), pp. 9-26.
- Cochrane, J. (2011) “Presidential address: Discount rates”, [*Journal of Finance* 66\(4\)](#), pp. 1047-1108.
- Daniel, K., Hirshleifer, D., and Sun, L. (2020) “Short-and long-horizon behavioral factors”, [*Review of Financial Studies* 33\(4\)](#), pp. 1673-1736.
- De Groot, W., Pang, J., and Swinkels, L. (2012) “The cross-section of stock returns in frontier emerging markets”, [*Journal of Empirical Finance* 19\(5\)](#), pp. 796-818.
- De Roon, F., Nijman, T., and Werker, B. (2001) “Testing for mean-variance spanning with short sales constraints and transaction costs: The case of emerging markets”, [*Journal of Finance*](#), 56(2), 721-742.
- Detzel, A. L., Novy-Marx, R., & Velikov, M. (2023) “Model selection with transaction costs”, [*Journal of Finance* 78\(3\)](#), pp. 1743-1775.
- Dong, X., Li, Y., Rapach, D. E., and Zhou, G. (2022), “Anomalies and the expected market return”, [*Journal of Finance*](#), 77(1), pp. 639-681.
- Eun, C. and Huang, W. (2007) “Asset pricing in China’s domestic stock markets: is there a logic?”, [*Pacific-Basin Finance Journal* 15\(5\)](#), pp. 452-480.
- Fama, E. and French, K. (1993) “Common risk factors in the returns on stocks and bonds”, [*Journal of Financial Economics* 33\(1\)](#), pp. 3-56.
- Fama, E. and French, K. (2015) “A five-factor asset pricing model”, [*Journal of Financial Economics* 116\(1\)](#), pp. 1-22.
- Fama, E. and French, K. (2018) “Choosing factors”, [*Journal of Financial Economics* 128\(2\)](#), pp. 234-252.
- Fama, E., and French, K. (2021) “The value premium”, [*Review of Asset Pricing Studies* 11\(1\)](#), pp. 105-121.
- Fletcher, J. (2018) “Bayesian tests of global factor models”, [*Journal of Empirical Finance* 48](#), pp. 279-289.
- Fletcher, J. (2019) “Model comparison tests of linear factor models in U.K. stock returns”, [*Finance Research Letters* 28](#), pp. 281-291.
- Gao, Y., Han, X., Li, Y. and Xiong, X. (2021) “Investor heterogeneity and momentum-based trading strategies in China.” [*International Review of Financial Analysis*](#), 74, 101654.

- Gao, Y., Zhao, W., and Wang, M. (2021) “The comparison study of liquidity measurements on the Chinese stock markets”, [Emerging Markets Finance and Trade](#), 58(2), pp. 483-511.
- Gibbons, M., Ross, S., and Shanken, J. (1989) “A test of the efficiency of a given portfolio”, [Econometrica](#) 57(5), pp. 1121-1152.
- Gu, M., Jiang, G., and Xu, B. (2019) “The role of analysts: An examination of the idiosyncratic volatility anomaly in the Chinese stock market”, [Journal of Empirical Finance](#) 52, pp.237-254.
- Guidolin, M., Hansen, E., and Lozano-Banda, M. (2018) “Portfolio performance of linear SDF models: an out-of-sample assessment”, [Quantitative Finance](#) 18(8), pp. 1425-1436.
- Guo, B., Zhang, W., Zhang, Y., and Zhang, H. (2017) “The five-factor asset pricing model tests for the Chinese stock market”, [Pacific-Basin Finance Journal](#) 43, pp. 84-106.
- Hanauer, M. (2020) “A comparison of global factor models”, [SSRN working paper](#).
- Hanauer, M., and Lauterbach, J. (2019) “The cross-section of emerging market stock returns”, [Emerging Markets Review](#) 38, pp. 265-286.
- Hansen, E. (2022) “Economic evaluation of asset pricing models under predictability”, [Journal of Empirical Finance](#) 68, pp. 50-66.
- Hou, K., Mo, H., Xue, C., and Zhang, L. (2019) “Which factors?” [Review of Finance](#) 23(1), pp. 1-35.
- Hou, K., Qiao, F., and Zhang, X. (2023) “Finding Anomalies in China”, [SSRN working paper](#).
- Hou, K., Xue, C., and Zhang, L. (2015) “Digesting anomalies: An investment approach”, [Review of Financial Studies](#) 28(3), pp. 650-705.
- Hou, K., Xue, C., and Zhang, L. (2020) “Replicating anomalies”, [Review of Financial Studies](#) 33(5), pp. 2019-2133.
- Hsu, J., Viswanathan, V. Wang, M., and Wool, P. (2018) “Anomalies in Chinese A-shares”, [Journal of Portfolio Management](#) 44(7), pp. 108-123.
- Hu, G., Chen, C., Shao, Y., and Wang, J. (2019) “Fama-French in China: size and value factors in Chinese stock returns”, [International Review of Finance](#) 19(1), pp. 3-44.
- Hu, G., Pan, J., and Wang, J. (2021) “Chinese capital market: an empirical overview”, [Critical Finance Review](#) 10(2), pp 125-206.
- Huang, Y., Yang, J., and Zhang, Y. (2013) “Value premium in the Chinese stock market: free lunch or paid lunch?”, [Applied Financial Economics](#) 23(4), pp. 315-324.
- Huang, T. (2019). “Is the Fama and French five-factor model robust in the Chinese stock market?” [Asia Pacific Management Review](#) 24(3), pp. 278-289.
- Huber, D., Jacobs, H., Müller, S., and Preissler, F. (2023) “International factor models”, [Journal of Banking and Finance](#) 150, 106819.

- Israel, R., Laursen, K., and Richardson, S. (2021) “Is (systematic) value investing dead?”, [*Journal of Portfolio Management*](#) 47 (2), pp. 38-62.
- Jacobs, H., and Müller, S. (2020) “Anomalies across the globe: Once public, no longer existent?”, [*Journal of Financial Economics*](#) 135(1), pp. 213-230.
- Jansen, M., Swinkels, L., and Zhou, W. (2021) “Anomalies in the China A-share market”, [*Pacific-Basin Finance Journal*](#), 68, 101607.
- Jansen, M., Swinkels, L., and Zhou, W. (2023) “More powerful tests for anomalies in the China A-share market”, *Journal of Portfolio Management* (forthcoming).
- Jiang, F., and Kim, K. (2020) “Corporate governance in China: a survey”, [*Review of Finance*](#) 24(4), pp. 733-772.
- Kan, R., Wang, X., and Zheng, X. (2022) “In-sample and out-of-sample Sharpe ratios of multi-factor asset pricing models”, [*SSRN working paper*](#).
- Karolyi, G. (2016) “Home bias, an academic puzzle”, [*Review of Finance*](#) 20(6), pp. 2049-2078.
- Ledoit, O., Wolf, M., and Zhao, Z. (2019) “Efficient sorting: a more powerful test for cross-sectional anomalies”, [*Journal of Financial Econometrics*](#) 17(4), pp. 645–686.
- Leippold, M., and Rueegg, R. (2018) “The mixed vs the integrated approach to style investing: Much ado about nothing?”, [*European Financial Management*](#) 24(5), pp. 829-855.
- Leippold, M., Wang, Q., and Zhou, W. (2022) “Machine-Learning in the Chinese stock market”, [*Journal of Financial Economics*](#), 145 (2), pp. 64-82.
- Lev, B., and Srivastava, A. (2022) “Explaining the recent failure of value investing”, [*Critical Finance Review*](#), 11(2), pp. 333-360.
- Liao, L., Liu, B., and Wang, H. (2014) “China’s secondary privatization: Perspectives from the split-share structure reform”, [*Journal of Financial Economics*](#) 113(3), pp. 500-518.
- Lin, Q. (2017) “Noisy prices and the Fama–French five-factor asset pricing model in China”, [*Emerging Markets Review*](#) 31, 141-163.
- Lin, Q. (2019) “Residual momentum and the cross-section of stock returns: Chinese evidence”, [*Finance Research Letters*](#) 29, pp. 206-215.
- Liu, J., Stambaugh, R., and Yuan, Y. (2019) “Size and value in China”, [*Journal of Financial Economics*](#) 134(1), pp. 48-69.
- Liu, Y., Shi, C., and Lian, X. (2019) “Refined book-to-market ratio and the cross-section of stock returns in China”, [*SSRN Working Paper*](#).
- Lu, X., Stambaugh, R., and Yuan, Y., (2017) “Anomalies abroad: Beyond data mining”, [*NBER Working Paper*](#).
- Lucey, B.M., Vigne, S.A., Ballester, L., Barbopoulos, L., Brzezczynski, J., Carchano, O., Dimic, N., Fernandez, V., Gogolin, F., González-Urteaga, A. and Goodell, J.W., Helbing, P., Ichev, R., Kearney F.,

- Laing, E., Larkin, C.J., Lindblad, A., Lončarski, I., Ly, K.C., Marinč, M., McGee, R.J., McGroarty, F., Neville, C., O'Hagan-Luff, M., Piljak, V., Sevic, A., Sheng, X., Stafylas, D., Urquhart, A., Versteeg, R., Vu, A.N., Wolfe, Simon., Yarovaya, L., Zaghini, A. (2018) "Future directions in international financial integration research - A crowdsourced perspective" [*International Review of Financial Analysis*](#), 55, pp. 35-49.
- Newey, W., and West, K. (1987) "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix", [*Econometrica*](#) 55(3), pp. 703-708.
- Novy-Marx, R. and Velikov, M. (2016) "A taxonomy of anomalies and their trading costs", [*Review of Financial Studies*](#), 29(1), pp. 104-147.
- Novy-Marx, R., and Velikov, M. (2019) "Comparing cost-mitigation techniques", [*Financial Analysts Journal*](#) 75(1), pp. 85-102.
- Pástor, L. (2000) "Portfolio selection and asset pricing models", [*Journal of Finance*](#) 55(1), pp. 179-223.
- Pástor, L., and Stambaugh, R. (2000) "Comparing asset pricing models: an investment perspective", [*Journal of Financial Economics*](#) 56(3), pp. 335-381.
- Shanken, J. (1987) "A Bayesian approach to testing portfolio efficiency", [*Journal of Financial Economics*](#) 19(2), pp. 195-215.
- Stambaugh, R. and Yuan, Y. (2017) "Mispricing factors", [*Review of Financial Studies*](#) 30(4), pp. 1270-1315.
- Soebhag, A., Van Vliet, B., and Verwijmeren, P. (2023) "Non-standard errors in asset pricing: Mind your sorts", [*SSRN Working Paper*](#).
- Tian, K. (2019) "Trading suspension bias in China stock market data", Working paper.
- Wang, F. and Xu, Y. (2004) "What determines Chinese stock returns?", [*Financial Analysts Journal*](#) 60(6), pp. 65-77.
- Wang, F., and Yan, X. S. (2021) "Downside risk and the performance of volatility-managed portfolios", [*Journal of Banking and Finance*](#) 131, 106198.
- Wang, Y. and Di Iorio, A. (2007) "The cross section of expected stock returns in the Chinese A-share market", [*Global Finance Journal*](#) 17(3), pp. 335-349.
- Weber, E., and Zhang, Y. (2012) "Common influences, spillover and integration in Chinese stock markets", [*Journal of Empirical Finance*](#) 19(3), pp. 382-394.
- Wolak, F. (1989) "Testing inequality constraints in linear econometric models", [*Journal of Econometrics*](#) 41(2), pp. 205-235.
- Xie, S., and Qu, Q. (2016). "The three-factor model and size and value premiums in China's stock market", [*Emerging Markets Finance and Trade*](#) 52(5), pp. 1092-1105.
- Xu, J., and Zhang, S. (2014) "The Fama–French three factors in the Chinese stock market", [*China Accounting and Finance Review*](#) 16(2), pp. 1-18.

Yin, L., and Yang, Z. (2022) “The profitability effect: Insight from a dynamic perspective.”, [*International Review of Financial Analysis*](#), 80, 102059.

Figure 1: Recursive model probabilities

This figure plots recursive model probabilities over time. Probabilities are calculated with Sharpe ratio multiple $\tau = 3$. Recursive probabilities begin in July 2000 and end each month between August 2001 through December 2019. Reported are the 7 best models from the 256 candidate models. The market factor is prejudged to be in each model. Probabilities are calculated using Chib, Zheng, and Zhao (2020) marginal likelihoods.

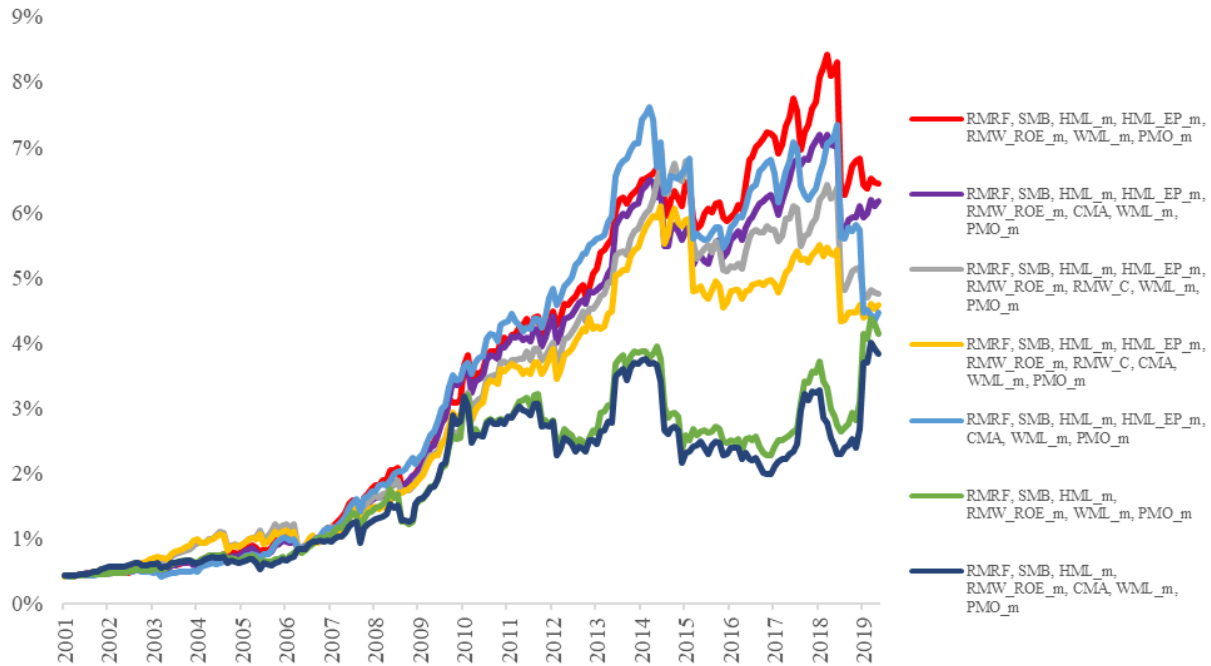


Figure 2: Cumulative factor probabilities

This figure plots (recursive) cumulative factor probabilities for 8 factors over time. Probabilities are calculated with Sharpe ratio multiple $\tau = 3$. Recursive probabilities begin in July 2000 and end each month between August 2001 through December 2019. The market factor is excluded because it is prejudged to be in each model. Probabilities are calculated using Chib, Zheng, and Zhao (2020) marginal likelihoods.

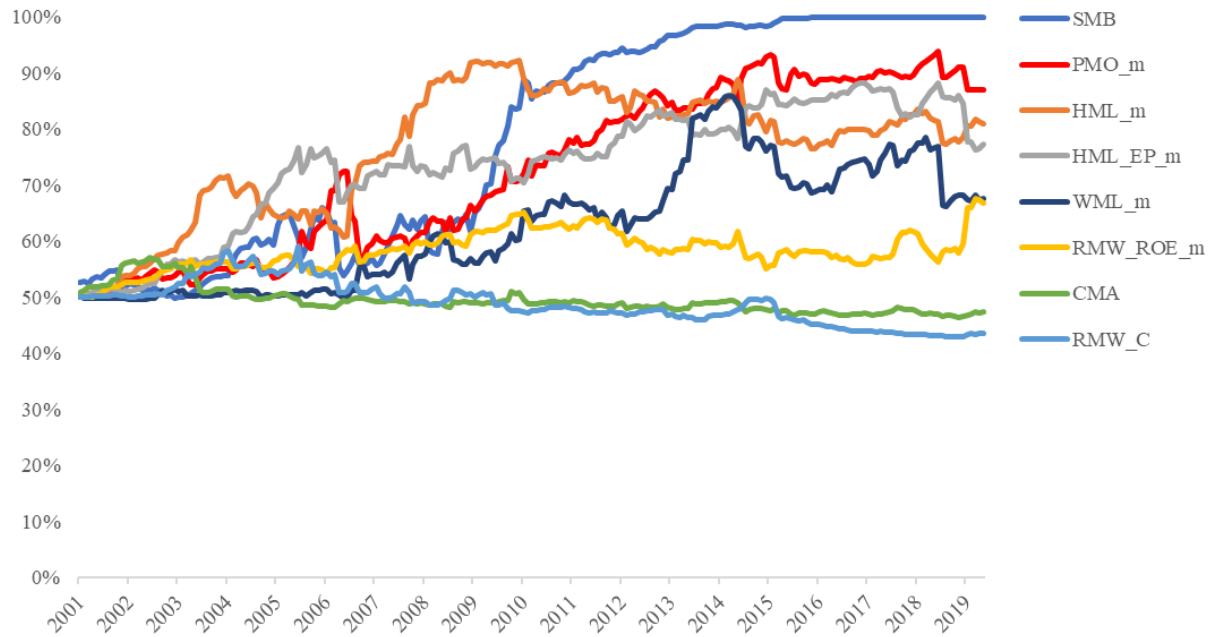


Table 1: Descriptive statistics of individual factor returns in the Chinese A-share market

This table reports the mean, standard deviation, t-statistic, and average turnover for the monthly factor as well as the CAPM alpha and t-statistic over the sample period July 2000 through December 2019. CAPM alphas are estimated through a regression of monthly factor returns on the excess returns of the value-weighted market portfolio. The definition of each of the factors is described in Section 2.2 and Appendix B. The t-statistics are calculated using Newey and West (1987) standard errors; values in bold are significant at the 5% significance level.

		Mean	Std dev	t-stat	Alpha	t-stat	Turnover
Market	RMRF	0.57	7.80	0.96			
Size	SMB	0.39	4.51	1.27	0.34	1.15	19.86%
Value	HML	0.34	4.02	1.37	0.32	1.33	8.27%
	HML_m	0.70	4.40	2.58	0.67	2.59	33.70%
	HML_EP_m	1.17	3.68	5.60	1.24	6.14	49.81%
Profitability	RMW_OP	0.04	2.30	0.32	0.06	0.48	7.52%
	RMW_CbOP	0.25	2.70	1.31	0.33	1.94	10.19%
	RMW_ROE_m	0.80	3.41	3.84	0.87	4.31	50.37%
	PERF_m	0.62	4.19	2.27	0.70	2.84	51.89%
Investment	CMA	0.06	2.28	0.39	0.05	0.34	11.56%
	MGMT_m	0.21	2.11	1.71	0.24	1.93	69.31%
Sentiment	WML_m	0.14	3.87	0.59	0.18	0.80	64.62%
	PMO_m	0.88	3.44	4.21	0.95	4.92	106.46%

Table 2: Correlation

This table reports the sample correlation coefficients between the factors over the sample period July 2000 through December 2019. The definition of each of the factors is described in Section 2.2 and Appendix B.

	RMRF	SMB	HML	HML_m	HML_EP_m	RMW_OP	RMW_CbOP	RMW_ROE_m	PERF_m	CMA	MGMT_m	WML_m	PMO_m
RMRF	1.00												
SMB	0.16	1.00											
HML	0.06	-0.42	1.00										
HML_m	0.06	-0.30	0.87	1.00									
HML_EP_m	-0.28	-0.72	0.39	0.33	1.00								
RMW_OP	-0.12	-0.24	0.03	-0.14	0.29	1.00							
RMW_CbOP	-0.40	-0.69	0.15	0.00	0.68	0.52	1.00						
RMW_ROE_m	-0.32	-0.62	-0.07	-0.20	0.79	0.39	0.75	1.00					
PERF_m	-0.30	-0.27	-0.48	-0.67	0.30	0.41	0.59	0.71	1.00				
CMA	0.05	0.20	0.47	0.49	-0.14	-0.24	-0.24	-0.47	-0.48	1.00			
MGMT_m	-0.16	0.02	0.41	0.49	0.09	-0.10	0.02	-0.18	-0.30	0.63	1.00		
WML_m	-0.15	-0.13	-0.27	-0.57	0.09	0.27	0.33	0.39	0.77	-0.14	-0.24	1.00	
PMO_m	-0.27	0.05	-0.24	-0.14	0.02	0.11	0.17	0.17	0.24	-0.11	0.09	0.06	1.00

Table 3: Sharpe ratios of popular asset pricing models

This table provides an overview of the factor models and their Sharpe ratios and turnover. Sharpe ratios are estimated over the sample period July 2000 through December 2019. Factor models are presented in order of increasing Sharpe ratios. *p*-values are reported for the null hypothesis that the squared Sharpe ratio is zero. Model turnover is the average monthly turnover of the corresponding tangency portfolio. The column numbers show the factor weight in the tangency portfolio and indicate which factors are included in the factor model. FF3, FF5, and FF6 are the Fama and French (1993, 2015, 2018) three-, five-, and six-factor models. FF6_CP substitutes the operating profitability factor RMW_OP in FF6 for the cash-based operating profitability factor RMW_CbOP; FF6_CPm is proposed by Barillas et al. (2020) and uses a monthly updated value factor HML_m instead of the yearly updated HML. FF6_ROE substitutes RMW_CbOP in FF6_CPm for RMW_ROE. SY4 is the Liu and Stambaugh (2017) mispricing model and Q4 is the Hou, Xue, and Zhang (2015) *q*-factor model. LSY3 and LSY4 are the China-specific three- and four-factor models by Liu, Stambaugh, and Yuan (2019).

	FF3	FF5	FF6	FF6_CP	SY4	FF6_CPm	Q4	LSY3	FF6_ROE	LSY4
RMRF	0.08	0.07	0.05	0.06	0.05	0.05	0.06	0.07	0.04	0.08
SMB	0.43	0.61	0.51	0.34	0.22	0.33	0.22	0.36	0.29	0.29
HML	0.49	0.74	0.67	0.22	0.29					
HML_m						0.37			0.27	
HML_EP_m								0.57		0.47
RMW_OP		0.24	0.03							
RMW_CbOP				0.48		0.35				
RMW_ROE_m							0.46		0.36	
PERF_m					0.33					
CMA		-0.66	-0.58	-0.15		-0.31	0.26		-0.08	
MGMT_m					0.11					
WML_m			0.32	0.04		0.22			0.12	
PMO_m										0.17
# factors	3	5	6	6	4	6	4	3	6	4
Developed for	US	US	US	US	US	US	US	CN	US	CN
Sharpe ratio	0.117	0.121	0.151	0.268	0.288	0.445	0.447	0.570	0.615	0.625
Squared Sharpe	0.014	0.015	0.023	0.072	0.083	0.198	0.200	0.324	0.379	0.390
p-value	0.097	0.124	0.068	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Turnover (%)	12.65%	11.96%	19.93%	13.85%	50.43%	24.70%	30.63%	35.67%	35.93%	47.00%

Table 4: Model comparison tests

This table reports model comparison tests between different factor models using pairwise differences in squared Sharpe ratios and multiple-model comparison tests of the squared Sharpe ratios. The sample period is July 2000 through December 2019. Models are presented in order of increasing squared Sharpe ratios. The top panel contains the difference in the bias-adjusted sample squared Sharpe ratio of the factor model contained in the column and the factor model in the row. The middle panel reports *p*-values for the null hypothesis that the difference in squared Sharpe ratios is zero. The bottom panel reports the value of the likelihood ratio test statistic and the corresponding *p*-value for the null hypothesis that none of the alternative models is superior to the benchmark factor model in the column. An overview of the factor models is provided in Table 3. CZZ denotes the data-driven seven-factor model introduced in Section 4. Statistical significance at the 5% level is indicated in bold.

	FF3	FF5	FF6	SY4	FF6_CP	FF6_CPm	Q4	LSY3	FF6_ROE	LSY4	CZZ
<i>Difference in squared Sharpe ratios</i>											
FF3		0.00	0.01	0.06	0.07	0.18	0.19	0.31	0.37	0.38	0.42
FF5			0.01	0.06	0.07	0.18	0.19	0.31	0.36	0.38	0.42
FF6				0.05	0.06	0.18	0.18	0.30	0.36	0.37	0.41
SY4					0.01	0.13	0.13	0.25	0.31	0.32	0.36
FF6_CP						0.12	0.12	0.24	0.30	0.31	0.35
FF6_CPm							0.00	0.13	0.18	0.19	0.23
Q4								0.12	0.18	0.19	0.23
LSY3									0.05	0.07	0.11
FF6_ROE										0.01	0.05
LSY4											0.04
<i>p-values</i>											
FF3		0.32	0.16	0.19	0.00	0.00	0.01	0.00	0.00	0.00	0.00
FF5			0.08	0.23	0.05	0.00	0.01	0.00	0.00	0.00	0.00
FF6				0.29	0.06	0.00	0.01	0.00	0.00	0.00	0.00
SY4					0.83	0.06	0.03	0.00	0.00	0.00	0.00
FF6_CP						0.00	0.07	0.00	0.00	0.00	0.00
FF6_CPm							0.99	0.14	0.01	0.03	0.00
Q4								0.04	0.00	0.00	0.00
LSY3									0.33	0.00	0.00
FF6_ROE										0.84	0.19
LSY4											0.03
<i>Multiple-model comparison tests</i>											
LR	20.62	21.62	23.02	13.33	17.43	7.99	10.77	0.94	1.73	0.00	0.00
p-value	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.36	0.29	0.75	0.75

Table 5: Squared out-of-sample Sharpe ratios

The table presents the annualized out-of-sample Sharpe ratios for the different investigated models for rolling estimation periods of 30, 60, and 120 months. First, we determine the tangency portfolio weights for the different models based on the different rolling windows. These factor weights are then applied to calculate portfolio returns in out-of-sample performance evaluations. The out-of-sample analysis is performed from January 2003 (July 2005, 2010) to December 2019.

	FF3	FF5	FF6	SY4	FF6_CP	FF6_CPm	Q4	LSY3	FF6_ROE	LSY4	CZZ
30	-0.01	-0.02	-0.03	-0.01	0.04	0.07	0.13	0.28	0.23	0.31	0.26
60	-0.02	-0.03	-0.03	-0.02	0.03	0.04	0.13	0.23	0.20	0.26	0.22
90	-0.02	-0.03	-0.04	-0.02	0.02	0.05	0.13	0.23	0.21	0.28	0.23
120	-0.03	-0.04	-0.05	-0.02	0.02	0.00	0.22	0.30	0.14	0.27	0.17
Max SR2	0.01	0.01	0.02	0.07	0.08	0.20	0.20	0.32	0.38	0.39	0.43

Table 6: Factor spanning tests

This table reports factor spanning tests for the non-market factors in the CZZ model. The sample period is July 2000 through December 2019. Each factor in the columns is regressed on the remaining factors. “coeff” is the estimated regression coefficient of the factor in the rows and “t-stat” is the associated t-statistic calculated using Newey and West (1987) standard errors. Alpha denotes the regression intercept and values in bold indicate significance at the 5% level.

	SMB		HML_m		HML_EP_m		RMW_ROE_m		WML_m		PMO_m	
	coeff	t-stat	coeff	t-stat	coeff	t-stat	coeff	t-stat	coeff	t-stat	coeff	t-stat
Alpha	1.39	5.90	0.49	3.28	0.18	1.75	0.18	1.74	0.51	2.02	0.75	3.14
RMRF	-0.01	-0.37	0.01	0.75	-0.02	-1.63	0.00	-0.01	-0.02	-0.60	-0.11	-2.54
SMB	-	-	-0.28	-5.78	0.00	0.06	-0.15	-5.19	-0.20	-3.25	0.17	1.88
HML_m	-0.53	-4.43	-	-	0.44	11.20	-0.41	-11.49	-0.66	-5.84	0.17	1.14
HML_EP_m	0.01	0.06	1.16	12.16	-	-	0.76	19.89	0.27	1.37	-0.48	-2.33
RMW_ROE_m	-0.92	-4.13	-1.33	-12.36	0.94	18.29	-	-	-0.13	-0.60	0.70	2.86
WML_m	-0.19	-2.72	-0.34	-5.55	0.05	1.31	-0.02	-0.59	-	-	-0.04	-0.35
PMO_m	0.14	2.00	0.08	1.06	-0.08	-2.05	0.09	2.41	-0.03	-0.35	-	-

Table 7: Net squared Sharpe ratios

This table reports gross and net squared Sharpe ratios and pairwise differences in net squared Sharpe ratios. Net squared Sharpe ratios are calculated with factor returns adjusted for transaction costs. The sample period is July 2000 through December 2019. Models are presented in order of increasing net squared Sharpe ratios. The top panel reports gross and net squared Sharpe ratios, the associated *p*-values, and the average monthly turnover of the corresponding tangency portfolio. The middle panel shows the difference in the bias-adjusted sample squared Sharpe ratio of the factor model contained in the column and the factor model in the row. The bottom panel reports the associated *p*-values. An overview of the factor models is provided in Table 3. CZZ denotes the data-driven seven-factor model introduced in Section 4. Statistical significance at the 5% level is indicated in bold.

	FF6	FF3	FF5	SY4	FF6_CP	Q4	FF6_CPm	FF6_ROE	CZZ	LSY4	LSY3
<i>Gross and net squared Sharpe ratios</i>											
Squared Sharpe	0.023	0.014	0.015	0.083	0.072	0.200	0.198	0.379	0.431	0.390	0.324
p-value	0.068	0.097	0.124	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Squared Sharpe (net)	0.001	0.003	0.005	0.010	0.044	0.051	0.057	0.105	0.131	0.142	0.145
p-value (net)	0.367	0.290	0.276	0.162	0.009	0.003	0.003	0.000	0.000	0.000	0.000
Turnover (%)	19.93%	12.65%	11.96%	50.43%	13.85%	30.63%	24.70%	35.93%	44.76%	47.00%	35.67%
Turnover (net, %)	14.85%	11.60%	11.98%	53.09%	16.04%	30.99%	17.27%	33.93%	39.02%	38.04%	35.41%
<i>Difference in net squared Sharpe ratios</i>											
FF6		0.00	0.00	0.01	0.04	0.05	0.06	0.10	0.13	0.14	0.14
FF3			0.00	0.01	0.04	0.05	0.05	0.10	0.13	0.14	0.14
FF5				0.01	0.04	0.05	0.05	0.10	0.13	0.14	0.14
SY4					0.03	0.04	0.05	0.10	0.12	0.13	0.14
FF6_CP						0.01	0.01	0.06	0.09	0.10	0.10
Q4							0.01	0.05	0.08	0.09	0.09
FF6_CPm								0.05	0.07	0.08	0.09
FF6_ROE									0.03	0.04	0.04
CZZ										0.01	0.01
LSY4											0.00
<i>p-values</i>											
FF6		0.41	0.65	0.75	0.10	0.22	0.05	0.02	0.01	0.01	0.01
FF3			0.27	0.79	0.01	0.19	0.09	0.02	0.01	0.01	0.01
FF5				0.84	0.14	0.25	0.07	0.02	0.02	0.01	0.01
SY4					0.39	0.23	0.26	0.06	0.03	0.02	0.02
FF6_CP						0.88	0.33	0.14	0.09	0.07	0.06
Q4							0.88	0.00	0.04	0.02	0.02
FF6_CPm								0.22	0.13	0.10	0.09
FF6_ROE									0.06	0.23	0.21
CZZ										0.84	0.91
LSY4											0.57

Appendix A

Figure A1: Recursive model probabilities with net factor returns

This figure plots recursive model probabilities over time. Model probabilities are calculated with factor returns adjusted for transaction costs. Probabilities are calculated with Sharpe ratio multiple $\tau = 3$. Recursive probabilities begin in July 2000 and end each month between August 2001 through December 2019. Reported are the 7 best models from the 256 candidate models. The market factor is prejudged to be in each model. Probabilities are calculated using Chib, Zheng, and Zhao (2020) marginal likelihoods.

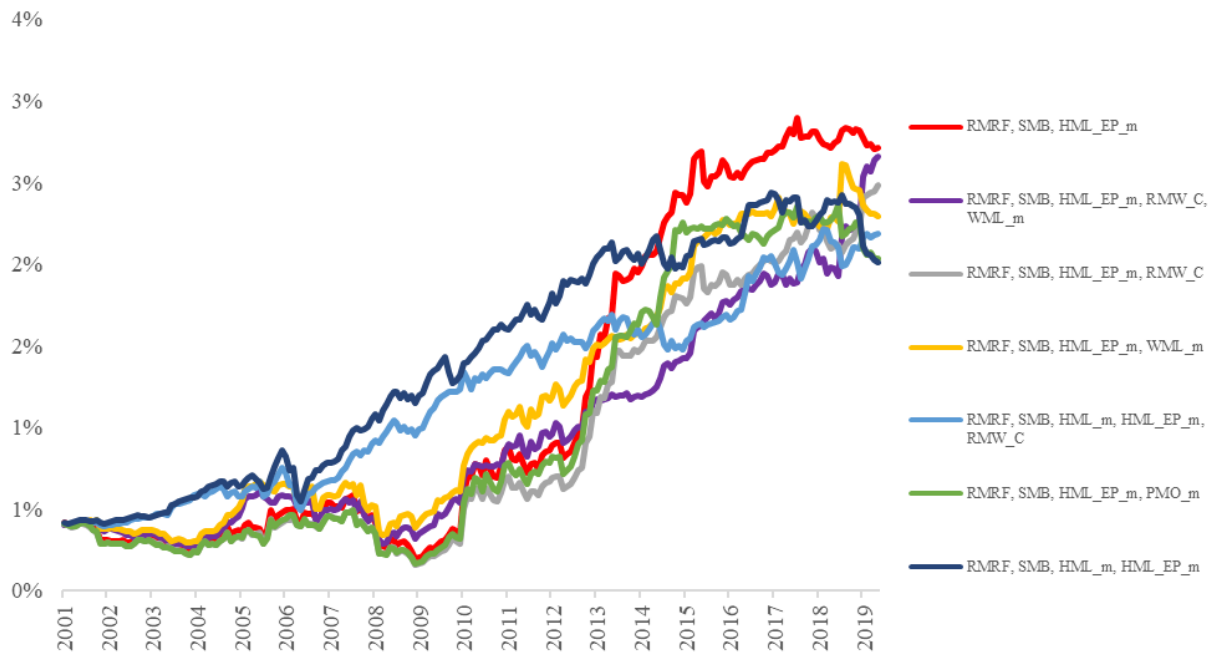


Figure A2: Cumulative factor probabilities with net factor returns

This figure plots (recursive) cumulative factor probabilities for 8 factors over time. Factor probabilities are calculated with factor returns adjusted for transaction costs. Probabilities are calculated with Sharpe ratio multiple $\tau = 3$. Recursive probabilities begin in July 2000 and end each month between August 2001 through December 2019. The market factor is excluded because it is prejudged to be in each model. Probabilities are calculated using Chib, Zheng, and Zhao (2020) marginal likelihoods.

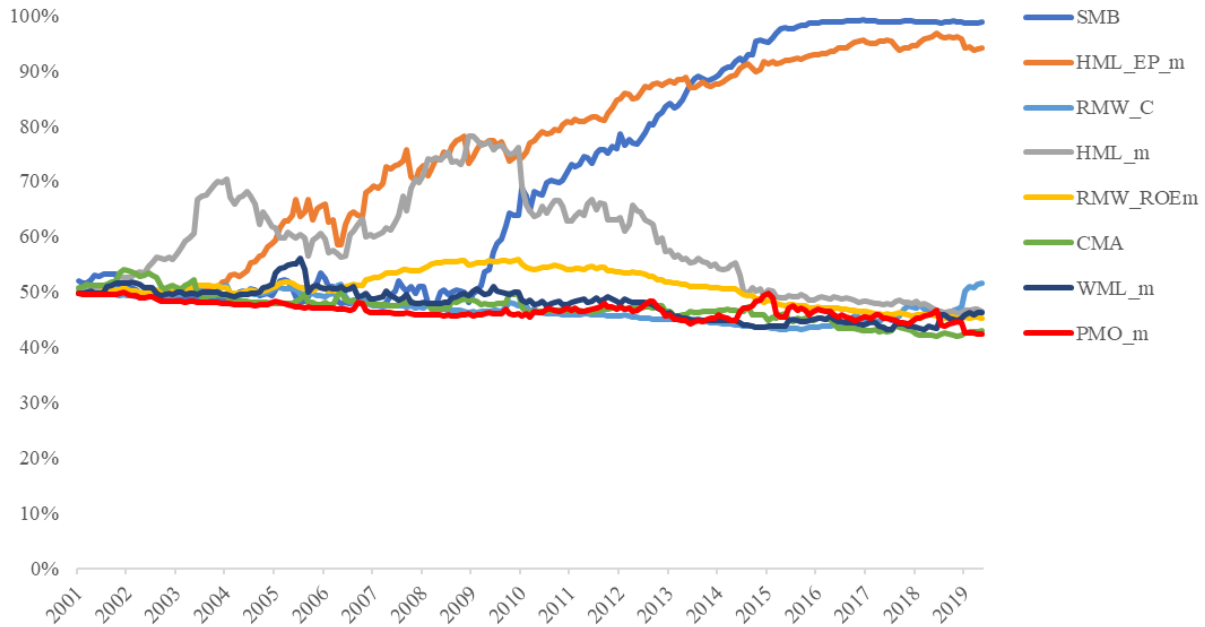


Table A1: Summary statistics for Chinese A-shares from 1990 to 2019

Reported are the number of listed firms (Firms), aggregate market capitalization of all A-shares (Market cap), average firm size (Firm size), the number of state-owned enterprises as a percentage of total listed firms (% of SOE), the percentage of non-tradable shares (% Non-tradable), and the return (Market return) and volatility (Market volatility) of a value-weighted market portfolio of all stocks. Market capitalization and firm size are in USD billions. All statistics except for return and volatility are year-end statistics. Returns are annual returns and volatility is the annualized monthly volatility.

Year	Firms	Market cap	Firm size	% of SOE	% Non-tradable	Market return	Market volatility
1990	6	0	0.07	66.7	70.03	-	-
1991	13	3	0.2	61.5	62.7	74.0	59.0
1992	53	19	0.4	52.8	80.2	145.1	108.2
1993	177	58	0.3	63.3	79.6	-13.0	72.8
1994	287	42	0.2	60.6	77.2	-28.6	133.6
1995	311	40	0.1	60.5	76.1	-13.4	37.0
1996	514	115	0.2	60.5	73.5	98.6	45.6
1997	719	206	0.3	61.8	71.9	26.6	31.8
1998	825	234	0.3	62.6	71.3	-5.1	22.4
1999	923	318	0.3	63.1	69.7	17.9	39.5
2000	1060	577	0.5	62.4	67.5	55.5	18.1
2001	1136	514	0.5	62.8	68.7	-23.4	19.2
2002	1199	455	0.4	63.1	68.8	-17.3	24.5
2003	1262	503	0.4	62.8	70.3	4.2	15.8
2004	1352	440	0.3	61.8	69.7	-15.5	20.4
2005	1356	394	0.3	61.8	68.5	-6.9	21.8
2006	1404	1129	0.8	61.1	73.1	134.4	28.6
2007	1526	4455	2.9	59.0	72.1	124.9	35.7
2008	1600	1772	1.1	57.2	63.2	-64.8	39.6
2009	1695	3561	2.1	55.0	38.4	91.0	33.1
2010	2039	3992	2.0	48.0	27.7	-7.4	24.9
2011	2318	3397	1.5	42.8	23.7	-24.3	16.4
2012	2470	3645	1.5	40.8	21.5	4.8	23.1
2013	2468	3902	1.6	40.7	16.8	3.8	21.1
2014	2591	6092	2.4	39.2	15.6	51.6	16.8
2015	2809	8172	2.9	36.8	21.8	32.3	40.1
2016	3031	7301	2.4	34.9	23.0	-12.3	32.2
2017	3466	8658	2.5	31.5	21.1	3.9	8.9
2018	3566	6325	1.8	31.0	18.9	-27.5	14.0
2019	3750	8487	2.3	30.0	18.7	29.1	20.1

Table A2: Factor returns with Main Board breakpoints and microcaps

This table reports the mean, standard deviation, and t-statistic for the monthly factor returns over the sample period July 2000 through December 2019. Size splits are made using Main Board breakpoints in the first set of columns. In the middle set, we include also microcaps, defined as the smallest 30% of stocks as measured by market capitalization. The final set of columns includes microcaps and mainboard breakpoints at the same time. The definition of each of the factors is described in Section 2.2 and Appendix B. The t-statistics are calculated using Newey and West (1987) standard errors; values in bold are significant at the 5% significance level.

		Mainboard breakpoints			Including microcaps			Mainboard breakpoints and microcaps		
		Mean	Std dev	t-stat	Mean	Std Dev	t-stat	Mean	Std Dev	t-stat
Market	RMRF	0.57	7.80	0.96	0.47	7.85	0.80	0.47	7.85	0.80
Size	SMB	0.31	4.77	0.95	0.67	4.82	2.03	0.66	4.83	1.98
Value	HML	0.34	4.03	1.39	0.33	3.65	1.50	0.32	3.69	1.41
	HML_m	0.69	4.41	2.57	0.71	3.99	3.02	0.70	4.01	2.93
	HML_EP_m	1.17	3.67	5.63	1.02	3.59	4.99	1.02	3.58	5.02
Profitability	RMW_OP	0.04	2.31	0.29	0.03	2.21	0.25	0.03	2.22	0.24
	RMW_CbOP	0.24	2.69	1.28	0.23	2.53	1.32	0.24	2.52	1.41
	RMW_ROE_m	0.80	3.41	3.86	0.54	3.21	2.73	0.55	3.20	2.78
	PERF_m	0.62	4.19	2.30	0.39	3.84	1.58	0.40	3.84	1.65
Investment	CMA	0.06	2.29	0.42	0.00	2.10	-0.02	0.00	2.11	-0.01
	MGMT_m	0.22	2.12	1.77	0.21	2.06	1.77	0.22	2.06	1.84
Sentiment	WML_m	0.14	3.87	0.60	0.07	3.60	0.29	0.07	3.60	0.32
	PMO_m	0.88	3.45	4.25	0.95	3.30	4.71	0.94	3.29	4.71

Table A3: Descriptive statistics of test asset returns

This table reports the mean, standard deviation, and t-statistic for the monthly test asset returns as well as the CAPM alpha and t-statistic over the sample period July 2000 through December 2019. The tests are the joint factor set analyzed in Hsu et al. (2018) and Liu, Stambaugh, and Yuan (2019) (25 factors) excluding those seven factors that are already included in one of the investigated factor models. The test assets are long-short factor returns constructed using 2×3 portfolios on size and the corresponding sorting characteristic. CAPM alphas are estimated through a regression of monthly factor returns on the excess returns of the value-weighted market portfolio. Detailed definitions of each of the test assets are described in Appendix C. The t-statistics are calculated using Newey and West (1987) standard errors; values in bold are significant at the 5% significance level.

		Mean	Std dev	t-stat	Alpha	t-stat
Value	Earnings-to-price ratio (annual)	0.37	3.66	1.62	0.43	1.98
	Sales-to-price ratio (annual)	0.31	3.52	1.47	0.29	1.44
	Cash flow-to-price ratio (annual)	0.37	2.25	2.70	0.38	2.79
	Dividend-to-price ratio (annual)	0.30	3.37	1.47	0.32	1.61
Profitability	Gross profitability (annual)	0.27	3.14	1.32	0.36	1.95
Accounting conservatism	Accrual (annual)	-0.07	1.98	-0.55	-0.05	-0.38
	Total accrual (annual)	0.17	2.39	1.08	0.20	1.33
	Net operating assets (annual)	0.44	1.91	3.42	0.43	3.52
Investment	Book value growth (annual)	-0.01	2.63	-0.07	-0.05	-0.31
Risk	Volatility (20 days)	0.65	3.94	3.03	0.77	3.65
	Maximum daily return (20 days)	0.71	3.15	4.02	0.79	4.73
	Volatility (250 days)	0.63	4.42	2.63	0.78	3.68
	Market beta (250 days)	0.27	3.83	1.25	0.41	2.09
	Idiosyncratic volatility (250 days)	0.54	3.65	2.68	0.62	3.36
Past returns	Short-term reversal (1 month)	0.96	3.71	4.26	0.94	4.15
	Long-term reversal (60 months)	0.31	3.38	1.47	0.28	1.40
Trading	Illiquidity (20 days)	0.98	3.53	4.69	1.00	4.97
	Turnover (250 days)	0.73	3.83	3.32	0.84	4.31

Table A4: Absolute pricing ability of asset pricing models for factors

This table reports the ability of factor models to price a cross-section of 30 factors over the sample period July 2000 through December 2019. The left-hand-side (LHS) assets in Panel A comprise the 18 additional anomalies reported in Hsu et al. (2018) and Liu, Stambaugh, and Yuan (2019) that are not already included in one of the investigated factor models (see Table 6 for an overview). Panel B also includes all right-hand-side (RHS) factors as test assets (see Table 2 for an overview). For each model, the RHS factors are excluded from the LHS test assets. The LHS assets are long-short factor returns constructed using 2×3 portfolios on size and the corresponding sorting characteristic. An overview of the factor models is provided in Table 3. GRS denotes the GRS statistic of Gibbons, Ross, and Shanken (1989) and p -value is the associated p -value. $A|a|$ is the average absolute regression intercept, $A|t|$ is the average absolute t -statistic of the regression intercepts, $\#\{|t| \geq 2\}$ is the number of absolute intercept t -statistics greater or equal to 2, and AR^2 is the average value of the regression R -squared. $Sh^2(f)$ is the maximum squared Sharpe ratio for the factor models and $Sh^2(a)$ is the maximum squared Sharpe ratio for the intercepts of the LHS assets.

	FF3	FF5	FF6	SY4	FF6_CP	FF6_CPm	Q4	LSY3	FF6_ROE	LSY4	CZZ
<i>Hsu et al. (2018) and Liu, Stambaugh, and Yuan (2019) anomalies (18 test assets)</i>											
GRS	5.22	5.18	5.67	6.50	5.62	4.05	3.38	2.69	3.21	2.12	2.80
p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.006	0.000
A a 	0.50	0.50	0.51	0.53	0.39	0.34	0.40	0.22	0.25	0.20	0.26
A t 	3.35	3.42	3.40	3.20	2.21	2.19	2.52	1.26	1.73	1.07	1.72
$\#\{ t \geq 2\}$	13	13	13	14	9	7	12	3	5	2	7
AR^2	0.46	0.52	0.54	0.46	0.52	0.56	0.47	0.36	0.57	0.41	0.59
$Sh^2(f)$	0.03	0.04	0.05	0.09	0.18	0.23	0.22	0.34	0.42	0.42	0.48
$Sh^2(a)$	0.45	0.46	0.51	0.60	0.57	0.43	0.35	0.31	0.39	0.26	0.36
<i>Hsu et al. (2018) and Liu, Stambaugh, and Yuan (2019) anomalies and RHS factors (30 test assets)</i>											
GRS	7.18	7.58	7.68	6.56	6.80	5.35	5.05	3.77	3.58	3.32	3.23
p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
A a 	0.55	0.57	0.58	0.55	0.44	0.40	0.43	0.24	0.25	0.19	0.23
A t 	3.57	3.88	4.00	3.43	2.94	2.59	2.58	1.22	1.78	0.95	1.64
$\#\{ t \geq 2\}$	19	20	20	21	15	12	16	5	8	2	8
AR^2	0.44	0.52	0.56	0.48	0.57	0.57	0.46	0.30	0.58	0.35	0.59
$Sh^2(f)$	0.03	0.04	0.05	0.09	0.11	0.23	0.22	0.34	0.42	0.42	0.48
$Sh^2(a)$	1.02	1.01	0.99	0.95	0.93	0.81	0.82	0.70	0.63	0.63	0.56

Table A5: Absolute pricing ability of asset pricing models for industries

This table reports the ability of factor models to price a cross-section of industry returns over the sample period July 2000 through December 2019. Because several industries have less than 10 stocks, we merged these with other industries. The final industry classification is then as follows: (1) 'Agriculture' and 'Mining', (2) 'Manufacturing', (3) 'Utilities' and 'Water management', (4) 'Construction' and 'Repair', (5) 'Wholesale and retail', 'Hospitality', and 'Leasing' (6) 'Transportation', (7) 'Software' and 'Culture and communication', (8) 'Financial', (9) 'Real estate', and (10) 'Science and Technical Research', 'Healthcare', 'Education', and 'Comprehensive'. GRS denotes the GRS statistic of Gibbons, Ross, and Shanken (1989) and p-value is the associated p-value. $A|a|$ is the average absolute regression intercept, $A|t|$ is the average absolute t-statistic of the regression intercepts, $\#\{|t| \geq 2\}$ is the number of absolute intercept t-statistics greater or equal to 2, and AR^2 is the average value of the regression R-squared. $Sh^2(f)$ is the maximum squared Sharpe ratio for the factor models and $Sh^2(a)$ is the maximum squared Sharpe ratio for the intercepts of the LHS assets.

	FF3	FF5	FF6	SY4	FF6_CP	FF6_CPm	Q4	LSY3	FF6_ROE	LSY4	CZZ
GRS	1.69	2.14	2.15	2.46	3.68	2.56	1.60	1.02	1.72	0.74	1.66
p-value	0.08	0.02	0.02	0.01	0.00	0.01	0.11	0.43	0.08	0.68	0.09
A a 	0.26	0.27	0.26	0.30	0.35	0.30	0.25	0.25	0.27	0.19	0.29
A t 	1.20	1.33	1.34	1.42	1.69	1.29	1.05	0.93	1.11	0.72	1.17
#{ t >=2}	2.0	2.0	3.0	3.0	4.0	1.0	2.0	0.0	2.0	0.0	1.0
AR²	0.85	0.87	0.87	0.87	0.86	0.87	0.85	0.84	0.86	0.84	0.86
Sh²(f)	0.03	0.04	0.05	0.11	0.09	0.23	0.22	0.34	0.42	0.42	0.48
Sh²(a)	0.08	0.10	0.10	0.13	0.18	0.14	0.09	0.06	0.11	0.05	0.11

Appendix B: Detailed factor definitions

Size

Firm size is calculated as the natural logarithm of a firm's end-of-month total A-share market capitalization.

Value

The value factors are calculated using either book-to-market ratios or earnings-to-price ratios. HML and HML_m are based on the book-to-market ratio, defined as book value over market value. Book value is defined as common shareholder's equity (total shareholder's equity minus the book value of preferred stocks) excluding minority interests. The book value of preferred stocks and minority interests are set to zero if missing. Market value refers to a proxy for end-of-month total market capitalization; the number of shares outstanding (including B- and H-shares) times the A-share's price. HML_EP_m is based on the earnings-to-price ratio, defined as total earnings over market value. Total earnings is defined as net profit excluding minority interest income (zero if missing).

HML is updated yearly at the end of June using the book value of fiscal year $t - 1$ and the market value measured at the end of December of year $t - 1$. HML_m is updated monthly using the book value of year $t - 1$ and the latest end-of-month market value. HML_EP_m is updated monthly using the most recently reported quarterly earnings and the latest end-of-month market value. We exclude firms with negative book values or earnings. To exclude stale earnings, firms only enter portfolio construction for HML_EP_m if the fiscal quarter of the latest available quarterly earnings is less than six months prior to portfolio formation. Prior to 2002, semi-annual earnings are used. The value factors are long firms with high valuation ratios.

Profitability

The profitability factors are calculated using operating profit-to-book (RMW_OP), cash-based operating profit-to-assets (RMW_CbOP), and return on equity (RMW_ROE_m). Operating profit-to-book is defined as operating profit minus interest expense (zero if missing), all divided by book value. Operating profit is sales minus cost of goods sold minus selling, general, and administrative expenses (SG&A, zero if missing). SG&A is the sum of business and management expenses,

selling expenses, and administrative expenses. We set each of these items of SG&A to zero if missing. We do not subtract research and development expenses since this item does not appear on Chinese income statements. Cash-based operating profit-to-asset is the sum of operating profit and the cash-based adjustment, divided by total assets. The cash-based adjustment by Ball, Gerakos, Linnainmaa, and Nikolaev (2016) is calculated using cash flow statement accruals rather than balance sheet items. The cash adjustment is defined as the sum of the decrease in accounts receivable, decrease in inventory, and increase in accounts payable. Return on equity is defined as quarterly earnings divided by one-quarter lagged book value, where earnings and book value are defined above.

RMW_OP and RMW_CbOP are updated yearly at the end of June of year t using the operating profit, book value, accruals, and assets from the fiscal year $t - 1$. RMW_ROE_m is updated monthly using the most recently reported quarterly earnings and the book value reported in the previous fiscal quarter. To exclude stale earnings, firms only enter portfolio construction for RMW_ROE_m if the fiscal quarter of the latest available quarterly earnings is less than six months prior to portfolio formation. Prior to 2002, semi-annual earnings are used. We exclude firms with negative earnings or negative book value from portfolio construction. The profitability factors are long firms with high profitability.

Investment

The investment factor (CMA) is based on the year-on-year change in assets, defined as total assets in the fiscal year $t - 1$ divided by the total assets in fiscal year $t - 2$, minus one. CMA is updated yearly at the end of June of year t and is long firms with low asset growth.

Momentum

The momentum factor (WML_m) is based on the stock return over the past year skipping the most recent month, calculated as the cumulative stock return from month $t - 12$ to month $t - 1$, where t is the rebalancing month. WML_m is updated monthly and is long stocks with high past returns.

Mispricing

The mispricing factors MGMT_m and PERF_m are composite factors which rank stocks based on their degree of underpricing. At the end of each month, stocks are ranked into percentiles for each

of the underlying variables. Lower ranks correspond to higher expected returns. *MGMT_m* is calculated as the average rank of asset growth, composite equity issuance (CEI), investment-to-assets (IA), net operating assets (NOA), and accruals. *PERF_m* is calculated as the average rank of gross profitability, stock momentum, and return on assets (ROA). For both factors, we require that at least three of the underlying variables are non-missing for a stock to be included in portfolio construction. Unlike Stambaugh and Yuan (2017), we do not include failure probability, Ohlson's O-score, and net stock issues (NSI). The former two variables are parameterized based on the U.S. stock market and are thus not relevant for China A-shares (cf. Lu, Stambaugh, and Yuan, 2017). NSI is excluded because CSMAR does not include a shares outstanding adjustment factor. *MGMT_m* and *PERF_m* are updated monthly and are long stocks with low ranks.

Asset growth: Asset growth is calculated as total assets in the fiscal year $t - 1$ divided by the total assets in fiscal year $t - 2$, minus one. Asset growth is updated yearly and firms with low asset growth have high expected returns.

CEI: CEI is calculated as the growth rate in the market capitalization not attributable to the total stock return, calculated as the growth rate in market value over the past twelve months minus the cumulative stock return over the past twelve months. CEI is updated monthly and firms with lower CEI have high expected returns.

IA: IA is calculated as the sum of the year-on-year changes in fixed assets and inventory, divided by one-year lagged total assets. IA is updated yearly in June using the fixed assets and inventory from fiscal year $t - 1$ and total assets from fiscal year $t - 2$. Firms with low IA have high expected returns.

NOA: NOA is calculated as net operating assets in year t scaled by total assets in year $t - 1$. The numerator is calculated as the difference between operating assets and operating liabilities. Operating assets is defined as total assets minus balance sheet cash, minus short-term investment. We require either balance sheet cash or short-term investment to be non-missing. Operating liabilities is total assets minus short-term loans, minus long-term loans, minus minority interest (zero if missing), minus common shareholder's equity excluding minority interest. We require either short-term loans or long-term loans to be non-missing. NOA is updated yearly in June using

balance sheet information from fiscal year $t - 1$ and firms with low NOA have high expected returns.

Accruals: Accruals are calculated as

$$ACC = 2 \times Accrual_t / (TA_{t-1} + TA_t)$$

$$Accrual_t = (\Delta CA_t - \Delta Cash_t) - (\Delta CL_t - \Delta STD_t - \Delta TP_t) - DP_t$$

where TA_t is total assets, $Cash_t$ is the balance of cash and cash equivalents, CL_t is current liabilities, STD_t is the sum of notes payable and long-term debt due within one year (zero if missing), TP_t is taxes payable (zero if missing), DP_t is the sum of depreciation of fixed assets, oil and gas assets, and bearer biological assets (zero if missing), and intangible asset amortization (zero if missing). Δ denotes the year-on-year difference and t denotes the year. Accruals is updated yearly using balance sheet and cash flow statement information from fiscal year $t - 1$ and firms with low accruals have high expected returns.

Gross profitability: Gross profitability is calculated as the ratio of gross profit to total assets. Gross profit is defined as sales minus cost of goods sold. Gross profitability is updated yearly using balance sheet and income statement information from fiscal year $t - 1$ and firms with high gross profitability have high expected returns.

Momentum: Stock momentum is calculated as the cumulative stock return from month $t - 12$ to month $t - 1$, where t is the rebalancing month. Stock momentum is updated monthly and firms with high past returns have high expected returns.

ROA: ROA is calculated as the ratio of quarterly earnings to one-quarter lagged total assets. Earnings is defined as net profit excluding minority interest income (zero if missing). ROA is updated monthly using the most recently reported quarterly earnings and the total assets reported in the previous fiscal quarter. To exclude stale earnings, firms only qualify for portfolio construction if the fiscal quarter of the latest available quarterly earnings is less than six months prior to portfolio formation. Prior to 2002, semi-annual earnings are used. We exclude firms with negative earnings. Firms with high ROA have high expected returns.

Sentiment

The sentiment factor (PMO_m) is based on abnormal trading turnover in the past month, where abnormal turnover proxies for sentiment. Abnormal turnover is defined as a stock's average daily share turnover over the past twenty trading days scaled by the average daily share turnover over the past 250 trading days. Daily share turnover is the ratio of the number of shares traded to the total number of shares outstanding. PMO_m is updated monthly and is long stocks with low abnormal trading turnover.

Appendix C: Test asset definitions

The test assets are constructed in the same way as the pricing factors, using 2×3 portfolio sorts on size and the corresponding sorting characteristic. Our set of test assets is based on the joint factor set studied by Hsu et al. (2018) and Liu, Stambaugh, and Yuan (2019), which spans 25 different factors. Since seven of these factors are already included in one of the investigated models, this leaves us with 18 remaining factors to price.

Value

Earnings-to-price: Earnings-to-price is calculated as the ratio of total earnings to price. Total earnings is defined as net profit excluding minority interest income (zero if missing). Price refers to a proxy of end-of-month total market capitalization; the number of shares outstanding (including B- and H-shares) times the A-share's price. We exclude firms with negative earnings. Earnings-to-price is updated yearly in June of year t using the earnings of fiscal year $t - 1$ and the market value measured at the end of December of year $t - 1$. The variable is long stocks with a low valuation ratio.

Sales-to-price: Sales-to-price is calculated as the ratio of total sales to price. Total sales is equal to operating revenue. Price refers to a proxy of end-of-month total market capitalization; the number of shares outstanding (including B- and H-shares) times the A-share's price. Sales-to-price is updated yearly in June of year t using the sales of fiscal year $t - 1$ and the market value measured at the end of December of year $t - 1$. The variable is long stocks with a low valuation ratio.

Dividend-to-price: Dividend-to-price is calculated as the ratio of total dividends to price. Total dividends is calculated as the total monetary value of dividends paid out to shareholders over the previous year. Price refers to a proxy of end-of-month total market capitalization; the number of shares outstanding (including B- and H-shares) times the A-share's price. We exclude firms that do not pay dividends. Dividend-to-price is updated yearly in June and is long stocks with a low valuation ratio.

Cash flow-to-price: Cash-flow-to-price is calculated as the ratio of total cash flow to price. Total cash flow is calculated as the net increase in cash or cash equivalents over the previous reporting period. Price refers to a proxy of end-of-month total market capitalization; the number of shares

outstanding (including B- and H-shares) times the A-share's price. We exclude firms with negative cash flows. Cash-to-price is updated yearly in June of year t using the cash flow of fiscal year $t - 1$ and the market value measured at the end of December of year $t - 1$. The variable is long stocks with a low valuation ratio.

Profitability

Gross profitability: Gross profitability is calculated as the ratio of gross profit to total assets. Gross profit is defined as operating revenue minus operating costs. Gross profitability is updated yearly in June of year t using the gross profit and total assets from the fiscal year $t - 1$ and is long stocks with high gross profitability.

Accounting conservatism

Accrual: Accrual are calculated as

$$ACC = 2 \times Accrual_t / (TA_{t-1} + TA_t)$$

$$Accrual_t = (\Delta CA_t - \Delta Cash_t) - (\Delta CL_t - \Delta STD_t - \Delta TP_t) - DP_t$$

where TA_t is total assets, $Cash_t$ is the balance of cash and cash equivalents, CL_t is current liabilities, STD_t is the sum of notes payable and long-term debt due within one year (zero if missing), TP_t is taxes payable (zero if missing), DP_t is the sum of depreciation of fixed assets, oil and gas assets, and bearer biological assets (zero if missing), and intangible asset amortization (zero if missing). Δ denotes the year-on-year difference and t denotes the year. Accruals is updated yearly using balance sheet and cash flow statement information from fiscal year $t - 1$ and is long firms with low accruals.

Total accrual: Total accrual is calculated as the difference between earnings in year t and operating cash flows in year t , scaled by total assets in year t . Total accrual is updated yearly using balance sheet and cash flow statement information from fiscal year $t - 1$ and is long firms with low accruals.

Net operating assets: Net operating assets is calculated as net operating assets in year t scaled by total assets in year $t-1$. The numerator is calculated as the difference between operating assets and operating liabilities. Operating assets is defined as total assets minus balance sheet cash, minus short-term investment. Operating liabilities represents total assets minus short-term loans, minus

long-term loans, minus minority interest, minus common shareholder's equity excluding minority interest. Net operating assets is updated yearly using balance sheet and income statement information from fiscal year $t - 1$ and is long firms with low net operating assets.

Investments

Book value growth: Book value growth is calculated as book value in the fiscal year $t - 1$ divided by the book value in fiscal year $t - 2$, minus one. Book value is defined as common shareholder's equity (total shareholder's equity minus the book value of preferred stocks) excluding minority interests. The book value of preferred stocks and minority interests are set to zero if missing. Book value growth is updated yearly and firms with low book value growth have high expected returns.

Risk

Volatility (20 days): One-month volatility is calculated as the standard deviation of daily stock returns over the past twenty trading days. One-month volatility is updated at the end of each month and is long stocks with low volatility.

Maximum daily return: Maximum daily return over the past twenty trading days equals the highest adjusted daily return over this period. Daily returns are adjusted in two ways. First, to account for a possible downward bias caused by trade suspensions, daily returns on suspended trading days are set equal to -99. Second, as suggested by Cheema, Nartea, and Man (2018), we aggregate the daily returns after a stock hits the upper price limit. Maximum daily return is updated at the end of each month and is long stocks with a maximum return.

Volatility (250 days): One-year volatility is calculated as the standard deviation of daily stock returns over the past 250 trading days. One-year volatility is updated at the end of each month and is long stocks with low volatility.

Market beta: Market beta proxies for systematic risk and is defined as the estimated slope coefficient from a regression of daily excess stock returns over the past 250 trading days on an intercept and the daily returns from the value-weighted market portfolio. The value-weighted market portfolio is calculated using all available A-share stocks. Market beta is updated at the end of each month and is long stocks with low beta.

Idiosyncratic volatility: Idiosyncratic volatility is calculated as the standard deviation of residuals from a regression of daily excess stock returns over the past 250 trading days on an intercept and the daily returns from the value-weighted market factor. Idiosyncratic volatility is updated at the end of each month and is long stocks with low volatility.

Past returns

Short-term reversal: Short-term reversal is calculated as the cumulative stock return over the past twenty trading days. Short-term reversal is updated at the end of each month and is long stock with low cumulative returns.

Long-term reversal: Long-term reversal is calculated as the cumulative monthly stock return over the past five years excluding the previous year, i.e., month $t = 60$ to month $t - 13$. Long-term reversal is updated at the end of each month and is long stocks with low cumulative returns.

Trading

Turnover: Turnover is calculated as the average of the daily share turnover over the past 250 trading days. Daily share turnover is the ratio of the number of shares traded to the total number of shares outstanding. Turnover is updated at the end of each month and is long stocks with low turnover.

Illiquidity: Illiquidity is calculated as the average stock illiquidity over the past twenty trading days. Stock illiquidity at day t is defined as

$$Illiq_t = |Ret_t|/Volume_t$$

where $|Ret_t|$ is the absolute value of the stock's daily return and $Volume_t$ is the dollar trading volume on day t , i.e., total number of shares traded times close price. Illiquidity is updated at the end of each month and is long stocks with high illiquidity.